

SI Appendix

The Metabolic Rate of the Biosphere and its Components

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1. Species-level biomass and energy

1.1. Source of data

Data on the biomass and metabolic rate of diverse species of organisms presented in Supplementary Dataset S1 were retrieved from the publications listed in Table S1. As described below, excluding microorganisms, the metabolic rates are grouped into basal, field or maximum rates. The largest number of data are available for basal metabolic rates. Note that many references with a large number of records include only one of these types of rates.

Metabolic rates are presented in this study in the power unit of Watts (Joule per second or J s^{-1} ; see Section 1.3). Metabolic rates are also presented per unit biomass, for example Watts per gram carbon (W (g C)^{-1}). This unit is called Mass-Specific Power (MSP).

Table S1 shows the measured type of metabolic rate (basal, field or maximum) and the studied groups of organisms, together with the number of records entered into the database. The total number of records in the database is ca. 10,500.

Figure S1 provides a graphical overview of the published sources of data in the database, showing the number of records for different groups of organisms included in Supplementary Dataset S1. For example, data for mammals originate primarily from [4] and [1], data for birds from [4], data for fishes from [10], while [9] and [3] present data for a broad spectrum of species across the phylogenetic tree.

Table S1. Contribution of different publications to the Supplementary Dataset S1, where full references are given.

Source	Rate Types	Groups of Organisms	# of Records
Savage et al. [1]	Basal	Mammals	1006
Bennett and Ruben [2]	Maximum	Mammals, Birds, Reptiles, Amphibians, Fishes,	6
Gillooly et al. [3]	Maximum	Mammals, Birds, Reptiles, Amphibians, Fishes	283
Hudson et al. [4]	Field	Mammals, Birds	1497
Nagy et al. [5]	Field	Mammals, Birds, Reptiles	164
Weibel et al. [6]	Maximum	Mammals	58
Lockyer [7]	Basal	Whales	24
Worthy et al. [8]	Basal	Killer whale (<i>Orcinus orca</i>)	4
Makarieva et al. [9]	Basal, Endogenous	Birds, Reptiles, Amphibians, Fishes, Cephalopods, Gelatinous Invertebrates, Decapods, Peracarida, Copepods and Krill, Insects, Tree Saplings, Seedlings, Heterotrophic Protozoa, Eukaryotic Microalgae, Cyanobacteria, Heterotrophic Prokaryotes	2833
Rubalcaba et al. [10]	Basal, Maximum	Fishes	4396
Makarieva et al. [11]	Endogenous, Growing (field)	Prokaryotes	104
Tijhuis et al. [12]	Maintenance	Prokaryotes	47
Schulz and de Beer [13]	Basal	Bacteria (<i>Thiomargarita namibiensis</i>)	1
Marschall et al. [14]	Maintenance	Bacteria (<i>Chlorobium</i> BS-1)	1
Scholten & Conrad [15]	Maintenance	Archaea (<i>Methanospirillum hungatei</i>)	2
Cordova et al. [16] ¹	Maximum	Bacteria (<i>Geobacillus</i> LC300, <i>Escherichia coli</i> MG1655)	2
Fuentes et al. [17] ¹	Maximum	Bacteria (<i>Escherichia coli</i> W3110)	1
Long et al. [18] ¹	Maximum	Bacteria (<i>Vibrio natriegens</i>)	1
Tang et al. [19] ¹	Maximum	Bacteria (<i>Geobacillus thermoglucosidasus</i>)	1
Wasylenko et al. [20] ¹	Maximum	Fungus (<i>Saccharomyces cerevisiae</i>)	1
Schönheit et al. [21] ²	Maximum	Archaea (<i>Methanobacterium thermoautotrophicum</i>)	1

¹Calculated from reported mass-specific substrate uptake rate assuming a ΔG for glucose or xylose oxidation of $-39.86 \text{ kJ (g C)}^{-1}$, equivalent to oxidation of 50 mM glucose at 0.2 bar O_2 .

²Calculated from reported specific growth rate and yield, with ΔG for methanogenesis calculated from reported culture conditions.

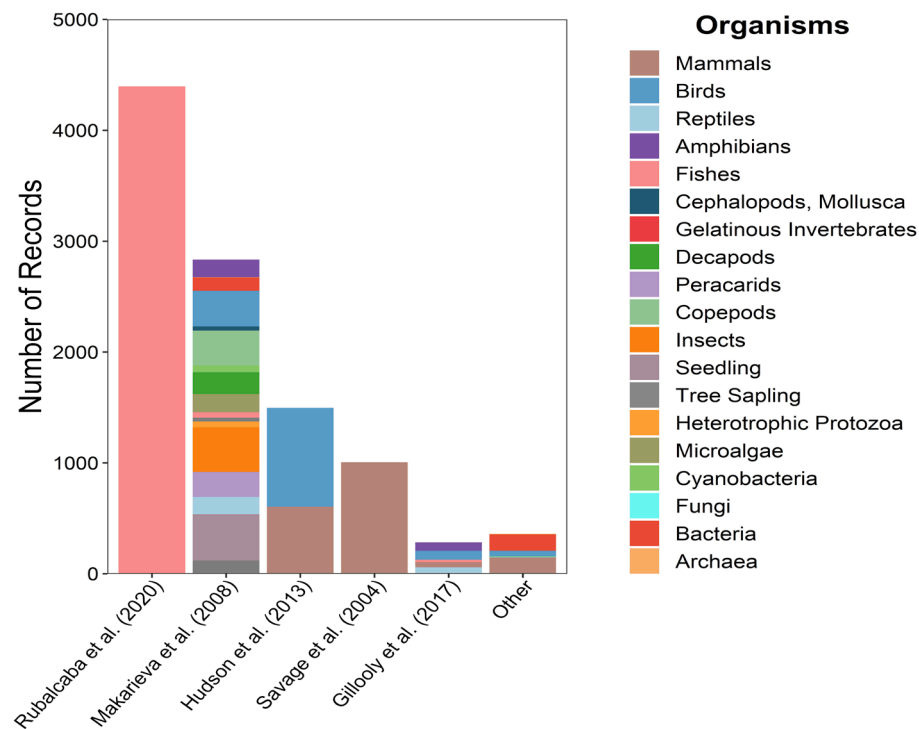


Figure S1. Total number of individual measurements that each publication contributed to the database, broken down by the group of organisms they contribute.

Studies of metabolic rates in individual species are highly selective towards taxonomic groups that are of particular scientific interest or that are particularly conducive to experimental measurements. The inclusion of published records in the database has also been partly selective for data that were already compiled by other authors for several species. It is striking that the largest number of records in the database are metabolic rates of fishes that were compiled by [10] (Fig. S2A). Some of these records represent multiple, independent measurements made on the same species, however, as discussed below. Second to the fishes are mammals and birds. This distribution shifts when comparing the number of unique species that have been studied (Fig. S2B). The largest diversity of species for which metabolic rates have been measured are found among the mammals, followed by birds, fishes, insects, reptiles and amphibians. There are also many different species of bacteria that have been studied. For some groups, such as seedlings and tree saplings, the number of records is considerable, but the diversity of species is very limited.

1.2 Basal, field and maximum metabolic rates

The types of metabolic rate described in publications have different names, depending on the group of organisms studied. We use the terms basal, field and maximum metabolic rate. The term “standard” metabolic rate is typically used for ectothermic animals in place of “basal”. The term “routine” metabolic rate is often used in place of “field”. We use the term “basal” for the minimum metabolic rate, typically measured in a controlled environment with little stimulation after fasting for a period, the duration of which depends on the type of organism. Maximum metabolic rates are measured in association with strenuous activity. For microorganisms such as bacteria, we consider “endogenous” rates to correspond to basal rates (following [9]) and “maximum” rates to be those

measured under optimal nutrient conditions in laboratory culture. Microorganism metabolic rates that are not clearly classifiable as endogenous or maximum rates are here considered “field” rates. For phototrophic organisms, we consider dark respiration rates to be basal rates.

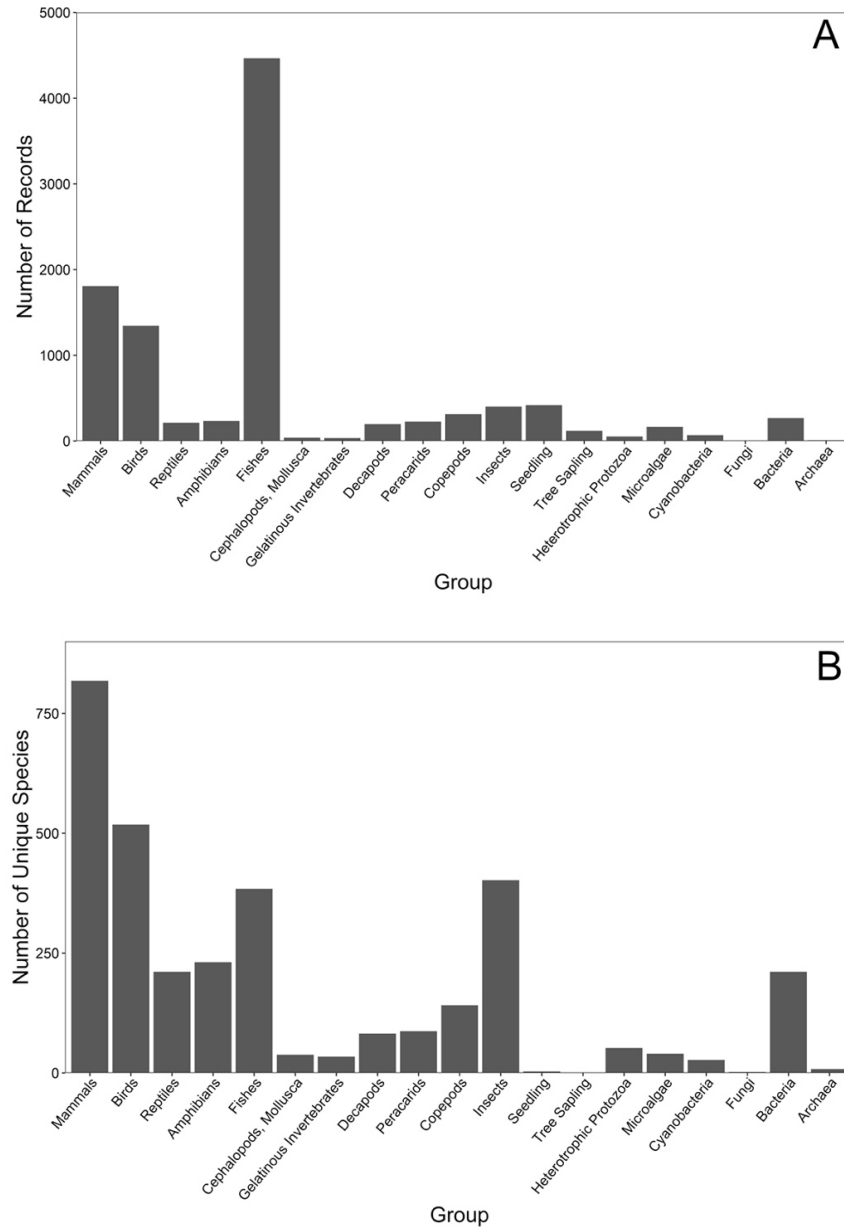


Figure S2. A) Number of records entered into the database for each taxonomic group. B) Number of unique species in the database for each taxonomic group.

For a limited number of cases, both basal and field and/or basal and maximum metabolic rates have been measured. On average for these species, field rates are about 3-fold higher than basal rates (Fig. S3A), while maximum rates are 5- to 10-fold higher (Fig. S3B). These averages include a few cases in which field or maximum rates fall below the corresponding basal rates, which clearly cannot be the reality for a given individual. This is likely a result of differing measurement conditions and/or natural variability in metabolic rates between different individuals, e.g., between

juveniles and adults. This same effect could produce anomalously high ratios, potentially including the few cases that approach a maximum-to-basal ratio near 100. For this reason, we view the mean and standard deviation (solid lines and shaded regions, respectively, in Fig. S3) as the preferable representation of dynamic range in Mass-Specific Power (MSP) over this span in body mass. This analysis does not include microorganisms: while it is common to characterize maximum growth rates in microorganisms, paired data that quantify the corresponding basal rates are uncommon. However, mean cell-specific rates that are 2-3 orders of magnitude below maximum rates have been observed in individual strains of both bacteria and archaea, including aerobes and anaerobes [22-23]. These measurements do not necessarily represent true basal rates for these strains, suggesting that the dynamic range for at least some strains of microorganisms may be larger still.

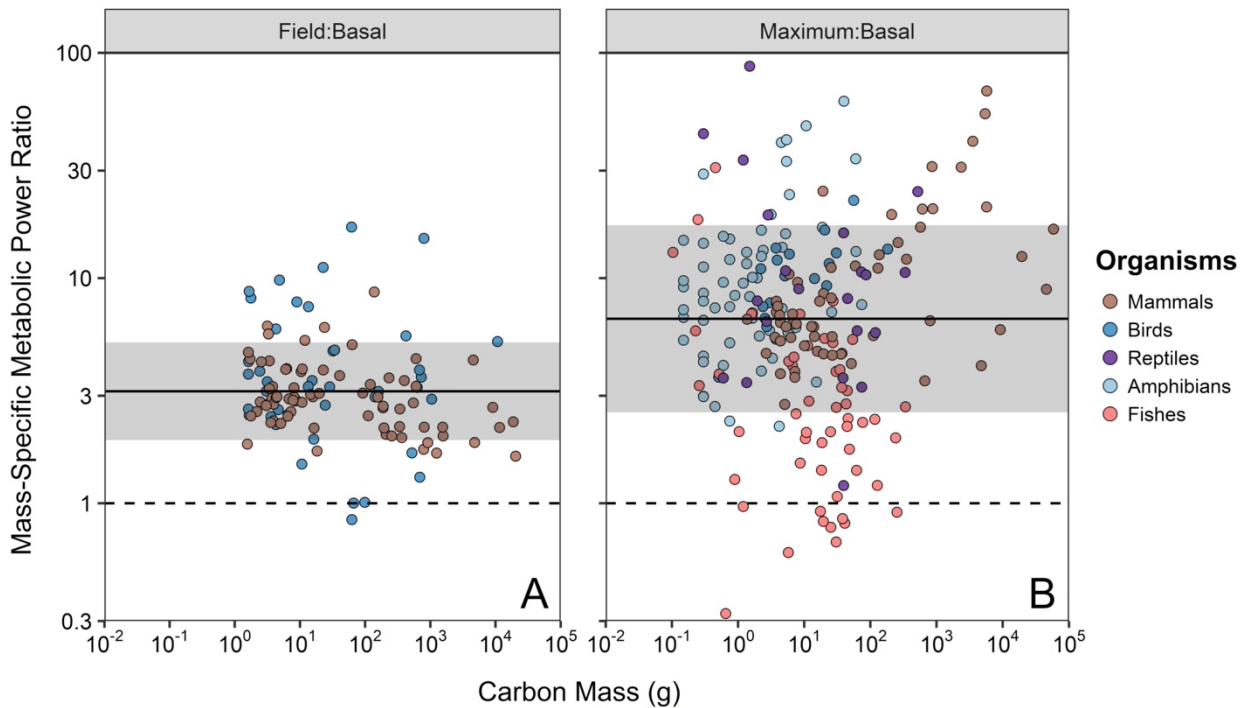


Figure S3. Ratios between A) field and basal MSP and B) maximum and basal MSP for individual metazoan species plotted against their mean carbon mass. The dashed lines and shaded regions are, respectively, the geometric means and standard deviations. The broken lines show the 1:1 ratio for the two metabolic parameters.

1.3. Measurement of metabolic rates

Measurements of metabolic rates are expressed in different units in the literature, depending on the type of organism and the method of measurement. Many studies on animals have used experimental measurements of O_2 respiration to determine metabolic (catabolic) rate. Data are generally expressed as mL O_2 consumed per unit time. This volumetric unit is converted to molar unit of O_2 using standard conditions of 0°C temperature and 1 bar pressure, whereby $1 \text{ mL } O_2 = 0.0446 \text{ mmol } O_2$.

Doubly isotope-labelled water was used in a number of studies to determine the field metabolic rate of organisms [24]. In this method, the organisms are provided with water in which the hydrogen and oxygen are partly replaced by ^2H (deuterium) and ^{18}O (e.g., [25]). It is then followed over time how fast the organism eliminates the two stable isotopes from the body. The loss rate of deuterium in bodily fluids shows how fast the organism eliminates the water isotopes via this route. The excess loss of ^{18}O in produced CO_2 is a measure of the respiration rate of organic carbon. The method provides data for respiration rate in terms of produced CO_2 .

The conversion between CO_2 and O_2 depends on the organic material being respired. The respiratory quotient, RQ, i.e. the molar ratio between CO_2 produced and O_2 consumed, may vary between 0.7 and 1.0, depending on the type of organic material respired. It is lowest for lipids (0.7) and highest for carbohydrates (1.0) with proteins in between. A common value used to represent a mixed diet is an RQ of 0.8 ($\Delta\text{CO}_2:\Delta\text{O}_2$) [26].

Metabolic rates are presented in the database, Supplementary Dataset S1, in the energy unit of Watts ($\text{W} = \text{J s}^{-1}$). For aerobic organisms, metabolic rates are mostly published in units of O_2 consumed. The conversion from oxygen respiration to energy units depends on the organic substrate respired and is ca. 19, 20 and 21 $\text{J (mL O}_2)^{-1}$ for proteins, lipids and carbohydrates, respectively [9]. We here use a mean value of 20 $\text{J (mL O}_2)^{-1}$.

Published metabolic rates reflect measurements made at a range of different temperatures and it is well documented that metabolic rates change with temperature. In Supplementary Dataset S1, we present rates at both the original measurement temperature and, to support comparison across species, at 25°C . The normalization of rates to 25°C uses the equation given by Makarieva et al. [9]:

$$W(25^\circ\text{C}) = W(T) * Q_{10}^{(25^\circ\text{C}-T)/10^\circ\text{C}} \quad (\text{Equation S1})$$

where W is the metabolic power, Q_{10} is the fold-change in metabolic power for a 10°C temperature increase, and T is the temperature of measurement. The Q_{10} values applied were generally 2.0, except for amphibians (2.21), molluscs (2.50), fishes (1.65), and reptiles (2.44) [9]. Temperature recalculation was not done for endothermic animals.

1.4. Conversion of mass units

Biomass data are given in different units in the literature: wet mass (WM), dry mass (DM) or organic carbon mass (C). Conversion factors between wet mass and dry mass are differentiated according to taxonomy in Supplementary Dataset S1. According to references cited in Supplementary Dataset S1, the ratio of dry mass to wet mass (DM/WM) used in our database ranges from 0.04 in the medusae of Cnidaria (jellyfish) to 0.40 in insects, with intermediate values of 0.26 in fishes, 0.3 in bacteria, 0.34 in birds and 0.38 in mammals. Makarieva et al. [9] used a conversion factor of $\text{DM/WM} = 0.3$ for all organisms. However, for some organisms, especially for jellyfish, the mass-specific metabolic rates are extremely low when calculated relative to wet mass but fall within the general trend of the other groups when calculated relative to dry mass, using the appropriate DM/WM conversion factor. In order of precedence, when available, the species-specific DM/WM conversion factor was used followed by the average or a crude intermediate value for a taxon-specific range. If neither value is known (in the cases of amphibians, reptiles, seedlings, and tree saplings), the average value of 0.3 used by Makarieva et al. [9] was applied.

Another important ratio is the organic carbon mass to dry mass ratio (C/DM), which is generally taken to be 0.5. The true ratio can vary across taxa based on differing contents of protein, carbohydrate, lipid etc. in the organisms. For example, Persson et al. [27] assumed the carbon content to be 45%, 48% and 50% of the dry weight of fungi, bacteria and soil animals, respectively. Elser et al. [28] estimated a C content of 46.4% in caterpillars. Measured data were in the range of 45-50% for bacteria [29] and 51-58% for cyanobacteria [30]. Following Bar-on et al. [31], we apply a uniform carbon-to-dry mass conversion factor of 0.5 for all organisms. The lone exception we make is in the specific case of humanity, for which detailed data are available. As described in Section 2, observational data support the use of a higher conversion factor (0.7) that is consistent with the higher carbon-to-dry mass ratio of fat and the relatively high fat content of the average human [32]. Higher carbon-to-dry mass ratios are likely also applicable in the case of other organisms with a high body fat percentage, such as marine mammals, livestock, and waterfowl. However, lacking direct data for these cases, we use the factor 0.5. Carbon mass-normalized metabolic and mass-specific rates in such cases may thus be slightly overestimated.

Data for fast-growing bacteria are mostly published without information on the cell-specific mass (e.g., [12]). In such cases, cell mass was either taken from separate literature or estimated using average cell dimensions for either the species or the genus (indicated by “species” or “genus”, respectively, in Supplementary Dataset S1). Cell dimensions were obtained primarily from Bergey’s Manual for Systematic Bacteriology [33] or the database BacDive (<https://bacdiv.dsmz.de/>). Organisms were excluded if the range between the minimum and maximum estimated cell mass was greater than an order of magnitude. Note that in these cases, substrate uptake data are generally reported on a mass-specific basis (e. g., per gram dry weight). For this reason, the MSP calculation is not affected by the lack of reported cell-specific mass.

1.5. Database structure

The complete database developed for this study is compiled as an Excel file (Supplementary Dataset S1).

The first worksheet in Supplementary Dataset S1 explains the column headings for all parameters entered into the database. It also gives full reference to all publications from which the data were retrieved and lists and explains the abbreviations used.

The second worksheet holds all the data available for ca. 10,500 recorded metabolic rate measurements. Each entry has been given an ID number. The different taxa have been affiliated to the relevant group of organisms (Table S2). For each organism, the database defines the taxonomic affiliation with respect to Domain, Kingdom, Phylum, Class, Order, Family, Genus and Species (Fig. S4). The taxonomy was assigned according to the Integrated Taxonomic Information system (ITIS, <https://itis.gov/>), which provides a taxonomic serial number (tsn), using the R package *ritis*. The table indicates the taxonomic rank from which the tsn data was obtained. In cases where species have changed names since the cited publication, it was still possible to find the correct tsn in ITIS. In ca. 1% of the cases, where species names had been misspelled or could not be found in ITIS, they were searched, and generally found, in the National Center for Biotechnology Information (NCBI) and the Encyclopedia of Life (EOL) databases.

The compiled database was screened to remove duplicate measurements. In many instances where species names had been misspelled or the taxonomy had been changed, there were multiple, identical entries in the published databases, or databases overlapped. This was in particular an issue with the database, FishBase, which is open to the public to enter new data (<https://www.fishbase.in/search.php>).

Table S2. Groups of organisms and their taxonomic level. Similar groups were used by Makarieva et al. [9].

Group of organisms	Level of taxonomy
Mammals	Class
Birds	Class
Reptiles	Class
Amphibians	Class
Fishes	Class
Cephalopods, Mollusca	Class, Phylum
Gelatinous invertebrates: Chaetognaths and medusae	Non-taxonomic group: Phylum and Subphylum
Decapods, Crustacea	Order, Subphylum
Peracarida, Crustacea	Suborder, Subphylum
Copepods and Krill, Crustacea	Subclass and Order, Subphylum
Insects	Class
Tree sapling	Non-taxonomic term
Seedling	Non-taxonomic term
Heterotrophic protozoa	Non-taxonomic group of a Subkingdom
Eukaryotic microalgae	Non-taxonomic group of a Phylum
Cyanobacteria	Phylum
Fungi	Kingdom
Bacteria and Archaea	Domains

For each entry in the database (Supplementary Dataset S1), the wet mass (WM) and dry mass (DM) of the studied organism are listed together with the DM/WM ratio. Dry mass is converted to carbon mass as described in Section 1.4. Metabolic rates are all converted to the unit of Watt (W) and include original rate data from the literature, corrected rates, temperature of measurement, and metabolic rate normalized to 25°C using the Q_{10} values described in Section 1.3. Metabolic rates have been grouped into basal, field or maximum rates, and the conversion factors used from mL O₂ consumed to Joule and to power in W (J s⁻¹) have been listed. Finally, full references and abbreviated reference codes are given, comments are added, and it is indicated whether data are from measurements on an individual organism or are averages of a number of measurements or individuals.

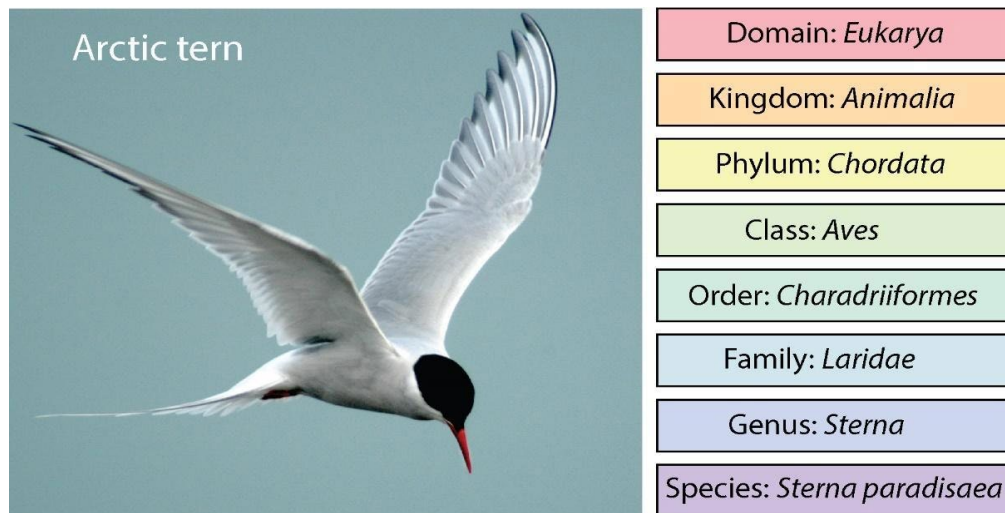


Figure S4. Taxonomic classification used in the database, here specified for an arctic tern (*Sterna paradisaea*; Photo: Bo Barker Jørgensen).

1.6. Interactive plot

All data in the database have been compiled, averaged by species (excluding seedlings and tree saplings), and are presented in a double-logarithmic, interactive plot (see “[Supplementary Interactive Plot](#)” or <https://doi.org/10.5281/zenodo.7877885>). This plot enables the user to toggle between the following options:

- 1) Type of data (toggles on left-hand side; by default, one of the options is always selected):
 Y-Axis: Metabolic rate (W) or Mass-specific metabolic rate ($W\ g^{-1}$)
 X-Axis: Wet mass (g), Dry mass (g) or Carbon mass (g)
 Temperature: Uncorrected (as measured and published) or normalized to 25°C.
- 2) Type of metabolic rate (on/off toggles at the top):
 Basal, Field and/or Maximum metabolic rate. By default, none of the three rate types is initially selected. For bacteria and archaea, endogenous rates toggle off/on with basal rates, maximum rates with maximum rates, and all other rates (e.g., maintenance or non-maximum growth) with field rates (see Section 1.2).
- 3) Taxonomic group of organisms (on/off toggles on right-hand side; each group is selected/deselected individually; the groups of organisms are the same as in Table S2; by default, all groups are selected).

For individual groups of organisms, a log-log-linear correlation line is shown for a) Metabolic rate versus Mass or b) Mass-specific metabolic rate versus Mass. Correlations are only shown for taxonomic groups that include 10 or more measurements and have a correlation coefficient >0.25 . The correlation lines are color-coded to match the corresponding data points and group identifier in the legend. When “Metabolic rate” is chosen, the plot shows, as a black line, the log-log-linear correlation with Mass for all the data. When “Mass-specific metabolic rate” is chosen, the plot shows the geometric mean and standard deviation (as fold-factor) for all data.

When hovering with the mouse above a data point in the plot, a small window shows the following parameters:

- a) “x” = mass in grams
- b) “y” = metabolic rate in Watts or mass-specific metabolic rate in Watts per gram
- c) “Species” = scientific species name
- d) “Group” = taxonomic affiliation of organism as specified in the list on the right-hand side
- e) “Type of rate” = basal, field or maximum metabolic rate for multi-cellular organisms and endogenous, growing or fast-growing for unicellular.

2. Biosphere-level mass

With the exceptions mentioned below and/or in the footnotes of main text Table 1, we use the mass estimates of Bar-on et al. [31]. Two of the biome categories considered in the present work bin together a number of individual taxa for which Bar-on et al. [31] developed individual estimates of mass and uncertainty (as reported in the Supplemental Materials of that paper). Specifically, ‘marine pelagic consumers’ encompasses fungi, animals and heterotrophic bacteria, archaea, and protists; and ‘terrestrial soil consumers’ encompasses primarily bacteria, archaea, fungi, and protists, along with smaller contributions from annelids and arthropods. To estimate the overall uncertainty in these bins, we adopted a similar strategy as Bar-On et al. [31]. Normal and/or log-normal random distributions were generated for each subgroup within a bin according to the subgroup’s mean and standard deviation (normal) or n-fold uncertainty (log-normal). The distribution for each bin was formed by the sum of the subgroup distributions. The mean and either standard deviation or n-fold uncertainty of the bin distribution is reported.

Estimate of mass and uncertainty for humanity: We base our estimate on the global population estimate of 8.0 billion in late 2022 (United Nations Food and Agriculture Organization (FAO); www.fao.org) and an estimated average human carbon mass of 15 ± 4 kg C. The latter number is based on West et al. [34], who calculated an average carbon mass for the US population, across all age groups, of 17.3 kg C with a range of 8-27 kg C. Applying the “range rule” yields an estimated standard deviation of ± 5 kg C, assuming a normal distribution in the data set. The average adult body mass in the US is higher than for the global population overall [32], so an estimate of average carbon mass for the global population must be scaled down from that calculated for the US population. The carbon mass calculations of West et al [34] use an average US adult total (wet) body mass of 72.7 kg, while Walpole et al. [32] estimate the average adult mass for the global population at 62.0 kg. We therefore scale the all-age US average carbon mass down by $62.0/72.7 = 0.853$ to obtain an estimate of all-age weighted average global carbon mass of 15 ± 4 kg C per capita, and thereby an estimated carbon mass for humanity of 0.12 Pg C. This estimate assumes a similar proportion of children to adults for the global population as in the US. It is estimated that the fraction of the population under 15 is presently 25% globally and 18% in North America (<https://data.worldbank.org/>), so the per-capita carbon mass estimate will overestimate the global value by ~10% if the average child under 15 has half the body mass of an average adult. Similarly, this approach to scaling assumes that carbon mass scales proportionately with wet body mass. However, adipose tissue has a relatively higher carbon content than protein (approx. 77% vs. 53%; [34]) so a US population that is proportionately more obese than the global average [32] would yield a somewhat higher carbon-to-total mass ratio. As a point of reference, if the 10 kg difference in average wet mass between the US and global populations were entirely attributable to adipose tissue, this would represent a difference of 7.7 kg C out of the US adult average of 20.3 kg C [34], or about 38%. This is an upper limit on the extent to which our chosen estimate of carbon mass may overestimate the global average because it does not factor in, e.g., differences in stature that may

also contribute to the difference in average mass. To factor in the considerations relating to US vs. global differences in obesity and the fraction of the population under 15 years of age, we revise the estimate of global average per capita carbon mass downward by 30% and the uncertainty upward by 30%, to 11 ± 5 kg C. For a global population of 8 billion, we therefore estimate the total carbon mass of humanity as 0.09 ± 0.04 Pg C. For comparison, Bar-on et al. [31] estimated the mass of humanity at 0.06 Pg C. The difference is attributable to differing assumptions for the dry-to-wet mass ratio (0.3 in Bar-on et al. [31], 0.4 in West et al. [34] - 33% higher in the latter case) and the carbon-to-dry mass ratio (0.5 in Bar-on et al. [31], 0.7 in West et al. [34] - 40% higher in the latter case), and to the 8% increase in global population since the time of that estimate.

Estimate of uncertainty for livestock: We use the mass estimate of Bar-on et al. [31], 0.1 Pg C, for livestock. While an associated estimate of uncertainty was not provided, the source data for average individual masses of the livestock types they considered provided uncertainty estimates for the masses of dairy cattle ($\pm 10\%$), other cattle ($\pm 25\%$), and swine ($\pm 20\%$) which, together, account for 80% of the total livestock mass in their estimate. From these individual uncertainties, we estimate an overall uncertainty of $\pm 15\%$, or ± 0.015 , based on propagation of errors in the weighted sum of these categories.

The total microbial biomass in subseafloor sediments at >0.1 meters below seafloor (mbsf) was taken from Kallmeyer et al. [35]. The authors compiled data for cell abundance in marine sediments and used a multiple correlation analysis of log-transformed cell counts to estimate a global subseafloor cell number of 2.9×10^{29} . Kallmeyer et al. [35] noted that a bootstrap exercise to check their analytical estimate of global cell abundance yielded a median value of 3.3×10^{29} cells, with the first SDs at 1.2×10^{29} and 8.0×10^{29} cells – which we approximate as reflecting a 2.5-fold uncertainty. Kallmeyer et al. [35] also determined mean cell sizes and calculated a mean cell biomass of $14 \text{ fg C cell}^{-1}$ with minimum and maximum estimates of 5 and $75 \text{ fg C cell}^{-1}$, respectively. Assuming that the minimum and maximum cell mass estimates represent the second standard deviations in log-normal distribution of cell masses implies a ~ 2 -fold uncertainty at one SD. Kallmeyer et al. [35] multiplied the mean cell mass by the estimated total number of cells in subseafloor sediments to yield an overall estimate of 4.1 Pg C for the global biomass beneath 0.1 m in seafloor sediments but did not provide an estimate of propagated uncertainty. To produce such an estimate, we conducted a Monte Carlo simulation in which cell mass was represented by a log-normal distribution ($N=5000$) with a median of 14 fg C and one SD at 2-fold, and total cell abundance was represented by a log-normal distribution ($N=5000$) with a median of 2.9×10^{29} cells and one SD at 2.5-fold. Computing the product of cell mass and cell abundance on a line-by-line basis yields a log-normally distributed population with one SD at 3.2-fold relative to the median, and we take this as approximating the uncertainty in the global biomass beneath 0.1 m in the subseafloor.

For surface sediments at 0-0.1 mbsf, we estimate total biomass, comprising benthic fauna and microbial components, as follows:

Rex et al. [36] and Wei et al. [37] estimated the $\log_{10}$ of the mean microbial cell abundance per m^2 in the global seabed to be 14.1 and 13.4, respectively. They found no correlation with ocean depth. The authors did not specify to which sediment depth these cell counts were calculated. However, we conclude from the papers that the data relate to the bioturbated zone, i.e. the top ca. 10 cm. Here, we use the data of Rex et al. [36] for a global calculation. In converting the log-transformed data of Rex et al. [36] for a global extrapolation, it should be considered that logarithmic transformation

tends to underestimate high values and overestimate low values. This skewness can be corrected for by the following correction factor: $e^{2.65 \times \text{variance}}$ [38]. We use the variance of the logarithmically transformed cell count data compiled by Kallmeyer et al. [35], which is 0.35. The correction factor is thus 2.5 and the resulting mean cell abundance of Rex et al. [36] is 3.1×10^{14} cells m^{-2} , or 3.1×10^{20} cells km^{-2} . For comparison, and to support an estimate of uncertainty, we made an estimate of global mean cell densities in the 0-0.1 m sediment interval from original data published by Kallmeyer et al. [35], which yielded a very similar abundance of 2.5×10^{20} cells km^{-2} . The 95% confidence interval of this overall mean value is $\pm 9\%$ or $\pm 0.23 \times 10^{20}$ cells km^{-2} . The sediments underlying ocean gyres stand out in the Kallmeyer et al. [35] dataset with cell abundances 100-1000-fold lower than in non-gyre sediments. D'Hondt et al. [39] estimate that gyre-associated sediments comprise 9%, at minimum, of the global ocean area. In our calculation of cell abundance of global seabed cell abundance, we therefore subtract 9% from the total ocean area to account for the gyre regions with lowest organic carbon oxidation rates.

The total area of the world ocean is 354×10^6 km^2 . Subtraction of 9% leaves 322×10^6 km^2 . The resulting global cell number in the top 0.1 m of sediments is then $3.1 \times 10^{20} \times 322 \times 10^6 = 1.0 \times 10^{29}$ cells. Thus, compared to the sub-seafloor, the upper 0.1 m of sediment harbors about one quarter of all microbial cells in the seabed.

The mean biomass of microbial cells in marine sediments has been estimated by several authors: 65 fg C cell^{-1} [40], 86 fg C cell^{-1} [41], 3 fg C cell^{-1} [36], 18 fg C cell^{-1} [42], 10 fg C cell^{-1} [37], 14 (5-75) fg C cell^{-1} [35], 36 fg C cell^{-1} [43], and 23 (19-31) fg C cell^{-1} [44]. When excluding the early (before 2008) extreme values, we calculate an overall mean of 20 fg C cell^{-1} and a standard deviation by this inter-comparison of ± 10 fg C cell^{-1} . For comparison, the mean cell-specific carbon content of bacteria is 7-40 fg C cell^{-1} in pelagic marine waters and 7-100 fg C cell^{-1} in coastal and nearshore marine waters [45].

Based on the very direct approaches used by Braun et al. [44] to determine cell carbon and cell volume, we will assume a mean microbial cell biomass of 23 fg C cell^{-1} for the top 0.1 m of the seabed. A similar number of 24 fg C cell^{-1} was used by Bar-On et al. [31] in their global estimate. The greater mean cell mass in the upper 0.1 m of sediments relative to that below (23 vs. 14 fg C) is consistent with the observation that energy-starved cells typically have smaller mass than those in more energy-replete environments [45]. Combining cell abundance and mean cell mass, the resulting global biomass of microbial cells in the top 0.1 m is then:

$$23 \times 10^{-15} \times 1.0 \times 10^{29} = 2.3 \text{ Pg C}$$

To estimate the overall uncertainty in this number, we conducted a Monte Carlo simulation in which cell mass was represented by a log-normal distribution ($N=5000$) with a median of 23 fg C and one SD at 2-fold, and total cell abundance was represented by a normal distribution ($N=5000$) with a mean of $1.0 (\pm 0.09) \times 10^{29}$ cells. Computing the product of cell mass and cell abundance on a line-by-line basis yields a log-normally distributed population with one SD at 2.1-fold relative to the median. We take this as approximating the uncertainty in the global biomass in the top 0.1 m of the seabed.

Previous estimates place the total mass of seabed fauna at 0.075 Pg C [37] and 0.040 Pg C [46]. The microbial biomass thus exceeds the faunal biomass in the upper 0.1 m of marine sediments by nearly two orders of magnitude.

3. Biosphere-level energy fluxes (power)

The vast majority of energy flow in the biosphere is inherently linked to carbon cycling, either via the photosynthetic capture and transformation of light energy during reduction of CO₂ into organic matter, or the oxidation of that organic matter back to CO₂ via aerobic or anaerobic respiration. Owing to the importance of these processes in the global CO₂ budget, extensive efforts have been made to quantify the associated carbon fluxes. We leverage this work as the basis for estimating energy fluxes to the different components of the biosphere. Specifically, we use estimates of the following fluxes, with values summarized in Table S3 and provide rationales for the choice of values described in the subsequent text. For all terrestrial C fluxes, we adopt the summary values given in the recent carbon cycle review by Crisp et al. (2022), with citations to the original source literature shown in Table S3. Marine C fluxes are drawn from a variety of literature sources.

The fluxes used in our calculations are:

Gross Primary Productivity (GPP) – The total flux of carbon fixed through photosynthetic reduction of CO₂ in a given system [47]. Here, we refer to estimates of GPP for both the terrestrial and marine domains.

Autotrophic Respiration (AR) – The rate of respiration of photosynthetically-fixed carbon by photosynthetic organisms in a given system. At the level of an individual organism, autotrophic respiration represents the partitioning of a portion of GPP into an oxidative process in order to produce ATP to fuel metabolic energy demands, finally resulting in oxidation of reduced carbon and release of CO₂. Here, we refer to estimates of Autotrophic Respiration for the terrestrial domain only. We are not aware of published estimates for global autotrophic respiration in the marine domain and, instead, estimate this quantity as the difference between marine GPP and NPP (see below).

Net Primary Productivity (NPP) – The net flux of organic carbon produced by photosynthetic organisms in a given system, following the partitioning of a fraction of GPP to autotrophic respiration. Thus, $NPP = GPP - AR$ [47]. Here, we refer to estimates of NPP for the marine domain, specifically. We do not use estimates of terrestrial NPP in our calculations and therefore do not include them in Table S3.

Heterotrophic Respiration (HR) – The rate of respiration of organic carbon by non-photosynthetic organisms in a given system. Here, we refer to estimates of R_H for both terrestrial soils and marine sediments.

Benthic Carbon Burial Flux ($J_{C\text{-burial}}$) – The fraction of seabed organic carbon, expressed as a flux, that escapes biological consumption to become “buried” over geological time.

These fluxes are dynamic in time and, in some cases, are subject to systematic change through time as a result of human-induced climate change. To the extent possible, the values used in this study are intended to represent a snapshot of present or near-present fluxes (e.g., the last few decades).

Table S3. Global carbon fluxes used to calculate biosphere energy fluxes.

	Abbreviation	Domain	Published Value	Converted Value (mol C·s ⁻¹)	References
Gross Primary Productivity	GPP	Terrestrial	115-190 Pg C yr ⁻¹	(3.0-5.0) × 10 ⁸	Cai and Prentice, 2020 ¹ [48]
Gross Primary Productivity	GPP	Marine	103-150 Pg C yr ⁻¹	(2.7-4.0) × 10 ⁸	Huang et al., 2021 [49]
Net Primary Productivity	NPP	Marine	52 ± 3.8 Pg C yr ⁻¹	(1.4 ± 0.1) × 10 ⁸	Silsbe et al., 2016 [50]
Autotrophic Respiration	AR	Terrestrial	64 ± 12 Pg C yr ⁻¹	(1.7 ± 0.3) × 10 ⁸	Ito, 2020 ¹ [51]
Heterotrophic Respiration	HR	Terrestrial (soils)	39 (33-46) Pg C yr ⁻¹	(1.3 ± 0.65) × 10 ⁸	Ciais et al., 2020 ¹ [52]
Heterotrophic Respiration	HR	Marine (sediments)	212 ± 55 Tmol C yr ⁻¹	(7 ± 1.7) × 10 ⁶	Jørgensen et al., 2022 [53]
Benthic Carbon Burial Flux	J _{C-burial}	Marine (sediments)	33 ± 25 Tmol C yr ⁻¹	(1.1 ± 0.8) × 10 ⁶	Multiple ²

¹As cited by Crisp et al. [47] as the source for the value given in their Table 4.

²Mean and propagated uncertainty among estimates from Wallman et al. [54], Sarmiento and Gruber [55], Dunne et al. [56], Regnier et al. [57].

3.1. Energy fluxes (Power) to Marine and Terrestrial Primary Producers

We calculate two distinct energy fluxes for the photosynthetic components of the biosphere. Species-level MSP for photosynthetic organisms is typically measured and reported as the dark respiration rate. The most comparable quantity that can be estimated at the biosphere level is the energy associated with autotrophic respiration, AR, and we report this value to support comparison between the organism- and biosphere-level data. Separately, we report ‘Photon Capture’ which encompasses the overall energy flux inherent in photoautotrophic carbon assimilation. We adopt this metric as representing the total capture and transformation of light energy by the photosynthetic components of the biosphere. To the extent that photosynthetic productivity fuels the vast majority of heterotrophic respiration on Earth, this metric also closely approximates the total power that supports the biosphere as a whole.

Photon Capture:

We estimate the power represented in global photosynthetic photon capture flux, P_{photon} , based on either NPP, as the quantity more commonly reported for marine systems, or GPP, as the quantity more commonly reported for terrestrial systems:

$$P_{\text{photon-marine}} = \text{NPP}_{\text{marine}} \cdot (1/\phi_{\text{max-marine}}) \cdot E_{\text{photon}}$$

$$P_{\text{photon-terrestrial}} = \text{GPP}_{\text{terrestrial}} \cdot (1/\phi_{\text{max-terrestrial}}) \cdot E_{\text{photon}}$$

Here, NPP and GPP are as described above. $\phi_{\max}$, the maximum quantum yield of carbon fixation ($\text{mol C (mol photons)}^{-1}$; [58]), equates gross or net photosynthetic carbon fixation to the photon flux that drives that process, while E_{photon} , the spectrum-weighted mean energy of photons in the photosynthetically active radiation (PAR) portion of the visible light spectrum, converts photon flux to energy flux. For both terrestrial and marine photosynthetic energy fluxes, we use $E_{\text{photon}} = 2.1 \times 10^5 \text{ J (mol quanta)}^{-1}$, converted from the factor $2.77 \times 10^{21} \text{ quanta s}^{-1} \text{ kW}^{-1}$ [59].

Quantum yield parameter, $\phi_{\max}$: A brief discussion of photosynthetic carbon fixation may be helpful in understanding what, specifically, our estimates of P_{photon} are intended to represent and how this underlies the choice of values for $\phi_{\max}$. In photosynthesis, absorbed light is used to generate both ATP and reducing power that support the “fixation” (reduction) of inorganic carbon (CO_2 , HCO_3^-) into triose phosphates via the Calvin-Benson-Bassham cycle, ultimately producing starch or sugar. Working at perfect efficiency, this process can support carbon fixation at a ratio of one mole C fixed per nine mol photons, so that $\phi = 0.111$ [58]. However, under environmentally relevant conditions, there are multiple processes that effectively drive ϕ to lower values (lower carbon fixation yield per quanta photons). These can be broadly grouped into (i) processes such as fluorescence and non-photochemical quenching (NPQ) that dissipate excess absorbed energy “upstream” of the electron transfer chain and (ii) “downstream” inefficiencies that affect the productivity of electron transfers, such as cyclic electron flow, in which the light absorbed gets used in recycling of electrons without yielding reduced product, and photorespiration, in which NADPH is not productive due to oxygenation of the carboxylating enzyme, RuBP, instead of productive carbon fixation. Our choice of ϕ values is intended to draw the line between these two sets of processes, and thereby distinguish absorbed energy from “captured” energy that leads to electron transfers. Absorbed photon energy that is dissipated upstream of the photosystems by fluorescence or NPQ is not included in our estimate of primary producer MSP. All “captured” photon energy – energy that successfully excites an electron from a chlorophyll molecule to provide reducing power, drive electron transfers, and the production of ATP – is included in the estimate. In this approach, the inefficiencies that lower the quantum yield of trapped photons are seen as inherent to carbon fixation under environmentally relevant conditions, and are thus accounted for in the choice of ϕ . The dissipation of absorbed energy as fluorescence or NPQ is minimized at low light levels, such that the quantum yield relative to absorbed light, still encompassing any downstream inefficiencies, is maximized [58]. We use the notation $\phi_{\max}$ to designate quantum yields that are applicable to such conditions and take this as the best approximation for excluding “upstream” effects while including “downstream” effects.

Choice of values for Marine Primary Producers:

For $\text{NPP}_{\text{marine}}$, we use $(1.4 \pm 0.1) \times 10^8 \text{ mol C s}^{-1}$, converted from $52.1 \pm 3.8 \text{ Pg C yr}^{-1}$ for the period 2002-2014 [50]. The uncertainty is the standard deviation among results in a sensitivity analysis conducted by Silsbe et al. [50] on eight model parameters. A more conservative take on uncertainty can be derived from Carr et al. [60], an intercomparison of 31 model-derived estimates of global marine NPP, representing five distinct modeling approaches. Estimates for global NPP during 1998 ranged from 32-79 Pg C yr^{-1} but the mean among all models was 50.7 Pg C yr^{-1} , similar to the value used here. Only one estimate in that study exceeded 70 Pg C yr^{-1} and 20 of 31 results fell in the range of 40-60 Pg C yr^{-1} . We adopt the NPP value of Silsbe et al. [50] as representative of refinements to modeling during the decade subsequent to the intercomparison study, but with a value that nevertheless approximates the mean observed in that intercomparison.

For $\phi_{\text{max-marine}}$ we use a value of $0.025 \pm 0.002 \text{ mol C (mol photons)}^{-1}$ from Silsbe et al. [50]. In that study, the value of ϕ_{max} specifically equates photon flux to NPP. It is parameterized as a linear function of light intensity with values ranging from 0.018 (high light) to 0.030 (low light), based on the observation that physiological acclimation to light regime causes variations in ϕ_{max} . The value of 0.025 ± 0.002 corresponds to the global mean and standard deviation in ϕ_{max} that results from this parameterization. We use this value, along with the value of $\text{NPP}_{\text{marine}}$ from the same study, for internal consistency in the calculation of P_{TP} . For comparison, Antoine and Morel [61] similarly estimated $\phi_{\text{max}} = 0.023 \pm 0.009$, while some studies have utilized globally constant (light-independent) values ranging from 0.030 [62] to as high as 0.060 [63-64]. See Silsbe et al. [50] for a more extensive discussion of ϕ_{max} in marine phytoplankton.

Choice of values for Terrestrial Primary Producers:

For $\text{GPP}_{\text{terrestrial}}$, we use $(3.0 - 5.0) \times 10^8 \text{ mol C s}^{-1}$, converted from 115-190 Pg C yr^{-1} ([47], based on [48]), estimated for 1982-2016. For comparison, Jung et al. [65] give a “currently most plausible range” of 110-150 Pg C yr^{-1} . We note that $\text{GPP}_{\text{terrestrial}}$ can have an interannual variability of 0.8-4.4 Pg C yr^{-1} around a mean that, during this period, had a trend and an uncertainty range much greater than the variability [66-67]. The global mean has also varied considerably through geologic time as the continents and plants have evolved [68].

For $\phi_{\text{max-terrestrial}}$, we use a value of $0.053 \pm 0.003 \text{ mol C (mol photons)}^{-1}$. The value represents the weighted mean and propagated standard deviation of observations made on C_3 plants (0.052 ± 0.003 , $N = 61$) and C_4 plants (0.057 ± 0.006 , $N = 56$) under low light conditions permissive to photorespiration ([58], and references therein). This calculation weights C_3 and C_4 plants at 75% and 25%, respectively, but the final value is relatively insensitive to this weighting. Note that Skillman [58] reports quantum yield values on a net carbon fixation basis, while we use these values to convert GPP to photon flux. However, corresponding measurements made under artificial conditions that inhibit photorespiration yield $\phi_{\text{max}} = 0.096$ ($\text{CV} = 15\%$), close to the theoretical maximum of 0.111. This indicates that $< 15\%$ of gross carbon fixation is consumed by autotrophic respiration under the experimental conditions. Since autotrophic respiration represents the difference between gross and net carbon fixation, this indicates that the reported quantum yield values (net carbon fixation basis) provide a close ($> 85\%$) approximation to the quantum yield of gross carbon fixation. We therefore consider the value of $0.053 \pm 0.003 \text{ mol C (mol photons)}^{-1}$ as an acceptable basis for equating GPP to photon flux.

In Table 1 of the main text, we report our estimates of power and MSP for terrestrial primary producers as a mean and standard deviation. Those values result from a Monte Carlo simulation that represents $\phi_{\text{max-terrestrial}}$ with a normal distribution and GPP with a flat distribution (equal weighting to all values within the indicated range).

Autotrophic Respiration:

We estimate the power represented in autotrophic respiration, P_{AR} , as:

$$P_{\text{AR}} = \text{AR}^* - \Delta G_{\text{AR}}$$

Here, ΔG_{AR} is the Gibbs energy change associated with autotrophic respiration. We use $\Delta G_{AR} = -4.74 \times 10^5 \text{ J (mol C)}^{-1}$, equivalent to the aerobic oxidation of glucose at 25°C under an atmosphere of 21% O_2 .

For the terrestrial domain we use $AR = (1.7 \pm 0.3) \times 10^8 \text{ mol C s}^{-1}$, converted from $64 \pm 12 \text{ Pg C yr}^{-1}$ [51].

For the marine domain we estimate AR as the difference between marine GPP and NPP, using the NPP_{marine} estimate given above and $GPP_{\text{marine}} = (3.4 \pm 0.4) \times 10^8 \text{ mol C s}^{-1}$, converted from $127 \pm 15 \text{ Pg C yr}^{-1}$. The latter value represents an estimate of the mean and standard deviation associated with a range of 103-150 Pg C yr^{-1} [49]. The estimation of standard deviation (SD) from range (R) uses the formula given by Ramirez and Cox [69]: $SD \approx R/(3*(\ln(N)^{0.5})-1.5)$, where N is the sample size. This formula assumes a normal distribution within the sample set. Lacking information on the exact sample size, N, underlying the range reported by Huang et al. [49], we set $N=10$ in order to produce a conservative estimate of standard deviation. As an alternative point of reference, simultaneous measurements of GPP and NPP in phytoplankton populations give GPP/NPP in the range of 2.7 for field-based measurements and 3.3 for culture-based measurements ([50], and references therein). Relative to our chosen estimate of $NPP_{\text{marine}} = 1.4 \times 10^8 \text{ mol C s}^{-1}$, the estimate of $GPP_{\text{marine}} = 3.4 \times 10^8 \text{ mol C s}^{-1}$ gives $GPP/NPP \approx 2.4$, slightly lower than the measured range.

3.2. Energy fluxes (Power) to Marine and Terrestrial Heterotrophic Consumers

We estimate energy fluxes to heterotrophic ‘consumers’ of organic matter as the product of globally integrated metabolic rate for a given biome in carbon-based units (e.g., mol C s^{-1}) and the Gibbs energy change (ΔG) for metabolism within that biome in carbon-based units (e.g., J (mol C)^{-1}).

Choice of values for ΔG :

Marine aerobic heterotrophic respiration. We assume that all respiration within the marine water column (pelagic consumers) and a portion of respiration within marine sediments occurs via aerobic respiration, with $\Delta G = -443,000 \text{ J (mol C)}^{-1}$. This value is derived from Table 1 of Lever et al. [45], which estimates the total Gibbs energy requirement for construction of a typical microbial cell ($5.1 \times 10^{-13} \text{ kJ cell}^{-1}$) as a weighted sum of the individual Gibbs energies of formation of the major cell constituent compounds. Table 1 of Lever et al. [45] also gives estimates of carbon atoms per cell in each of the compounds used for calculating the Gibbs energy change for biosynthesis. Summing the carbon atom estimates gives a corresponding value of $6.95 \times 10^8 \text{ C atoms cell}^{-1} = 1.15 \times 10^{-15} \text{ mol C cell}^{-1}$. Dividing the biosynthetic cost of $5.1 \times 10^{-13} \text{ kJ cell}^{-1}$ by the cell carbon content of $1.15 \times 10^{-15} \text{ mol C cell}^{-1}$ gives $-443 \text{ kJ (mol C)}^{-1} = -443,000 \text{ J (mol C)}^{-1}$. Using this value implies that marine NPP has a composition equivalent to that of bacterial cells. We consider this a useful approximation to the starting composition of marine NPP in that the bulk of that NPP derives from overturn of the (microbial) phytoplankton community. However, to the extent that subsequent consumption of NPP will preferentially remove some compounds over others, that composition will change with time. Even so, the associated ΔG for aerobic respiration will change only modestly, given that the aerobic consumption of a wide range of organic carbon compounds occurs with a relatively narrow range of ΔG . For comparison, Bradley et al. [70] represented aerobic respiration of organic carbon in marine sediments as the reaction of acetate (630 μM) with O_2 (80 μM) at pH 7.78 and 2 mM HCO_3^- and calculated $\Delta G = -34.72 \text{ kJ (g C)}^{-1} = -416,640 \text{ J (mol C)}^{-1}$.

Marine anaerobic heterotrophic respiration. We assume that anaerobic respiration of organic matter in marine sediments is dominated by sulfate reduction. Egger et al. [71] estimated that methanogenesis accounts for only 3-4 percent of the global anaerobic mineralization of marine sediment organic matter. For the Gibbs energy change associated with sulfate reduction, we adopt the value of $-32,200 \text{ J (mol C)}^{-1}$ used by Bradley et al. [70] ($-2.6833 \text{ kJ (g C)}^{-1}$; their Table S2, ‘nominal’ case). Note that, in our usage, $\Delta G = -32,200 \text{ J (mol C)}^{-1}$ is applicable to the complete oxidation of organic matter to CO_2 with sulfate – that is, we assume this energy yield to be shared by sulfate reducers and by the organisms that first degrade and ferment organic matter into substrates (e.g., H_2 and organic acids) that can be utilized by sulfate reducers. In this sense, it differs from the usage of Bradley et al. [70] who calculated this energy yield as specific to sulfate reducers only, based on the reaction of acetate ($630 \text{ }\mu\text{M}$) with sulfate (15.4 mM) to bicarbonate (25 mM), and bisulfide ($1 \text{ }\mu\text{M}$).

Terrestrial heterotrophic respiration. We assume that all consumption of terrestrial NPP occurs via aerobic respiration, with $\Delta G = -483,000 \pm 23,000 \text{ J (mol C)}^{-1}$. This value is the mean and standard deviation for gross heat of combustion among 35 plant/agricultural residues for which carbon content was also determined [72]. The values in that publication were reported for O_2 at unit fugacity; here, we adjust the value to an oxygen fugacity of 0.2 atm . In applying this value of ΔG (a value applicable specifically to aerobic respiration), we assume that the fraction of terrestrial NPP that is respired anaerobically, e.g., in inundated soils, is negligible. Assuming methanogenesis to be the primary mode of anaerobic carbon respiration in such soils allows this fraction to be estimated: An upper limit can be placed on the net consumption of terrestrial NPP via methanogenesis by assuming that the global methane flux of 520 Tg C yr^{-1} (CarbonTracker- CH_4 Project, NOAA ESRL, <http://gml.noaa.gov/ccgg/carbontracker-ch4/>) is entirely attributable to terrestrial ecosystems. The stoichiometry of organic C to methane during methanogenesis is approximately 2 to 1: $2[\text{CH}_2\text{O}] \rightarrow \text{CH}_4 + \text{CO}_2$ (here, “[CH_2O]” is meant to approximate organic matter). Accordingly, a methane flux of $520 \text{ Tg CH}_4 \text{ yr}^{-1}$ implies an upper limit on methanogenic consumption of terrestrial NPP of 1.4 Pg C yr^{-1} , or $<3\%$ of the total terrestrial NPP of $\sim 50 \text{ Pg C yr}^{-1}$ [52].

Estimates of globally integrated process rates:

Marine heterotrophic respiration, pelagic (water column). We estimate the total rate of heterotrophic respiration by marine pelagic consumers as $(1.3 \pm 0.1) \times 10^8 \text{ mol C s}^{-1}$. This value represents the difference between $\text{NPP}_{\text{marine}}$ and $\text{J}_{\text{C-benthic}}$, where the latter is the globally integrated flux of organic carbon to the seabed, with uncertainty reflecting the propagated error. For $\text{NPP}_{\text{marine}}$, we use $(1.38 \pm 0.10) \times 10^8 \text{ mol C s}^{-1}$, converted from $52.1 \pm 3.8 \text{ Pg C yr}^{-1}$, as described above. For $\text{J}_{\text{C-benthic}}$, we use $(7.9 \pm 1.9) \times 10^6 \text{ mol C s}^{-1}$, converted from $250 \pm 60 \text{ Tmol C yr}^{-1}$. The latter value represents the sum and propagated error of estimates of (i) the rate of global seabed oxidation of organic carbon with O_2 as the ultimate oxidant, (ii) the rate of global seabed oxidation of organic carbon with sulfate as the ultimate oxidant, and (iii) the burial flux of organic carbon. For (i), we use an estimate of $212 \pm 55 \text{ Tmol C yr}^{-1}$ [53]. This value results from comprehensive compilation and modelling of sediment O_2 uptake data from publications and available databases, which yielded a global sediment oxygen uptake flux of $255 \text{ Tmol O}_2 \text{ yr}^{-1}$. This estimate does not include ocean margin systems such as salt marshes, mangroves, seagrass beds or coral reefs. The O_2 uptake flux was converted to a carbon oxidation rate by assuming an average $\text{C}_{\text{org}}:\text{O}_2$ stoichiometry of 0.83 [53]. For reference, two recent studies estimated (i) as $139 \text{ Tmol C yr}^{-1}$ [73] and $99 \text{ Tmol C yr}^{-1}$ [46]. For (ii), we use a value of $7.6 \text{ Tmol C yr}^{-1}$. The estimate is based on a global marine sulfate reduction of $45 \text{ Tmol SO}_4^{2-} \text{ yr}^{-1}$, of which an estimated 90% is already included in the total O_2

uptake due to direct or indirect re-oxidation of the produced H_2S back to SO_4^{2-} [53]. This leaves $4.5 \text{ Tmol SO}_4^{2-} \text{ yr}^{-1}$, which corresponds to $7.6 \text{ Tmol C yr}^{-1}$ given a stoichiometry of C:S of 1.7:1 [53]. For (iii), we use a value of $33 \pm 25 \text{ Tmol C yr}^{-1}$, which is the mean and standard deviation among estimates of carbon burial flux from Wallman et al. [54], Sarmiento and Gruber [55], Dunne et al. [56] and Regnier et al. [57].

Marine heterotrophic respiration, sediments. For each of the two sediment depth intervals considered, 0-0.1 mbsf and >0.1 mbsf, we make separate estimates of the rates of carbon oxidation with O_2 vs. with sulfate as the ultimate electron acceptor. Use of the term “ultimate” here acknowledges that, although carbon oxidation can occur in sediments via multiple pathways, O_2 and sulfate dominate as the final sinks for organic matter electrons. Anaerobic carbon metabolism in marine sediments can occur via either sulfate reduction or (in subsurface sediments depleted of sulfate) methanogenesis, in addition to smaller contributions from reduction of nitrate, Mn(IV) and Fe(III). It is estimated that methanogenesis accounts for only 3-4 percent of the global mineralization of seafloor sediment organic matter [71]. Most of the methane produced diffuses upwards to become oxidized to CO_2 at the sulfate-methane transition [74]. The net chemistry of methanogenesis from organic matter ($2[\text{CH}_2\text{O}] \rightarrow \text{CH}_4 + \text{CO}_2$) followed by oxidation of methane with sulfate ($\text{CH}_4 + \text{SO}_4^{2-} + 2\text{H}^+ \rightarrow \text{CO}_2 + \text{H}_2\text{S} + 2\text{H}_2\text{O}$) is equivalent in both stoichiometric and thermodynamic terms to the oxidation of organic matter via sulfate reduction ($2[\text{CH}_2\text{O}] + \text{SO}_4^{2-} + 2\text{H}^+ \rightarrow 2\text{CO}_2 + \text{H}_2\text{S} + 2\text{H}_2\text{O}$), although the associated energy is distributed among differing groups of microorganisms. Thus, estimates of the rate of carbon oxidation with sulfate as the ultimate electron acceptor encompass both direct sulfate reduction and methanogenesis followed by anaerobic oxidation of methane by sulfate, and thereby the large majority of anaerobic carbon degradation in the seabed. The situation is similar in the case of O_2 as the ultimate electron acceptor, which may encompass (i) direct consumption of organic carbon in aerobic respiration (approx.: $[\text{CH}_2\text{O}] + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$), (ii) consumption of organic carbon via sulfate reduction (approx.: $[\text{CH}_2\text{O}] + \frac{1}{2} \text{SO}_4^{2-} + \text{H}^+ \rightarrow \text{CO}_2 + \frac{1}{2} \text{H}_2\text{S} + \text{H}_2\text{O}$), and (iii) the product H_2S subsequently consumed by oxidation with O_2 ($\frac{1}{2} \text{H}_2\text{S} + \text{O}_2 \rightarrow \frac{1}{2} \text{SO}_4^{2-} + \text{H}^+$). Stoichiometrically, and in terms of overall Gibbs energy change, the sum of reactions (ii) and (iii) is equivalent to reaction (i), but with the overall energy yield distributed among fermenters, sulfate reducers, and sulfide oxidizers rather than only among the organisms that mediate direct aerobic respiration.

Marine heterotrophic respiration, sediments 0-0.1mbsf. As described above, we estimate the rate of carbon oxidation with O_2 as the ultimate electron acceptor as $212 \pm 55 \text{ Tmol C yr}^{-1}$ [53], corresponding to $6.7 \pm 1.7 \times 10^6 \text{ mol C s}^{-1}$. The fraction of this flux that is attributable to seabed animals (benthic macro- and meiofauna) may be 25-40% of the total oxygen uptake in coastal sediments [75-77] but drops significantly across the shelf and into the deep sea [53, 78]. Much of the fauna-mediated oxygen uptake is due to bioirrigation, which provides O_2 not only to the animals but also to the microbial communities around their burrows.

We estimate the rate of carbon oxidation with sulfate as the ultimate electron acceptor as $0.49 \times 10^6 \text{ mol C s}^{-1}$, based on estimates of total global sulfate reduction in continental shelf sediments ($30 \text{ Tmol SO}_4^{2-} \text{ yr}^{-1}$), and in sediments beyond the shelf, at >200 m water depth ($15 \text{ Tmol SO}_4^{2-} \text{ yr}^{-1}$) [74, 79]. Ocean margin systems such as salt marshes, mangroves, intertidal flats and seagrass beds ($20 \text{ Tmol SO}_4^{2-} \text{ yr}^{-1}$) are here considered part of the terrestrial biosphere. Of the ($30+15=$) $45 \text{ Tmol SO}_4^{2-} \text{ yr}^{-1}$, only a fraction takes place in the top 0.1 m of the seabed. High-resolution data that discriminate sulfate reduction in the bioturbated top 0.1 m of the seabed relative to the total sediment column are scarce. On the continental shelf in the Baltic Sea to North Sea transition and

on the continental slope off Namibia, this fraction was found to vary widely between <10% and 90%, with a median value of 25-50% [80-81]. While most of the global shelf and slope is less eutrophic than these regions, they contribute correspondingly less to the global sulfate reduction rates. We here estimate that about 25% of the total, depth-integrated sulfate reduction takes place in the top 0.1 m in shelf sediments and about 10% at >200 m water depth. The global sulfate reduction in the top 0.1 m of the seabed is thus $(30 \times 0.25 + 15 \times 0.10 =) 9.0 \text{ Tmol SO}_4^{2-} \text{ yr}^{-1}$. With an estimated $\text{C}_{\text{org}}:\text{SO}_4^{2-}$ stoichiometry of 1.7:1 [74], this corresponds to $15.3 \text{ Tmol C yr}^{-1}$ or $0.49 \times 10^6 \text{ mol C s}^{-1}$. Given the ΔG values associated with carbon oxidation via O_2 ($-443 \text{ kJ (mol C)}^{-1}$) and sulfate ($-32.2 \text{ kJ (mol C)}^{-1}$), the total power in sediments <0.1 m comprises contributions of $(6.7 \pm 1.7) \times 10^6 \times 443 \times 10^3 / 10^9 = 3,000 \pm 800 \text{ GW}$ from carbon oxidation with O_2 , and $(0.49 \times 10^6 \times 32.2 \times 10^3 / 10^9 =) 16 \text{ GW}$ from carbon oxidation with sulfate. Thus, the estimated anaerobic carbon oxidation by sulfate reduction contributes only 0.5% compared to the energy flux by aerobic respiration and does not change significantly the rounded number of aerobic power.

Marine sediment consumers, >0.1 mbsf. We estimate the rate of carbon oxidation with O_2 as the ultimate electron acceptor as follows. In general, the rate of O_2 consumption in ocean margin, continental shelf, and continental slope sediments is sufficiently high that O_2 is depleted within millimeters to centimeters of the sediment-water interface. Organic carbon oxidation below 0.1 m in all such sediments occurs via anaerobic processes. However, in ocean regions deeper than 2,000-4,000 m, where the dissolved oxygen uptake rate is in the range of $0.1\text{-}5 \text{ mmol O}_2 \text{ m}^{-2} \text{ d}^{-1}$, the O_2 penetration depth may exceed 0.1 m (data compiled in Jørgensen et al., [53]). As an upper limit on the power associated with the penetration of O_2 beneath 0.1 mbsf, we use a value of 20.3 GW, which was calculated by Bradley et al. [70] calculated as the power associated with O_2 consumption for global “subseafloor” sediments. Their estimate derives from modeling of the decrease with depth of particulate organic carbon (POC) content in sediments. The depth-dependent decrease in POC content is generally restricted to sediments beneath the bioturbated surface layer (generally the upper 5-10 cm of sediment) because bioturbation, the churning of sediments by benthic faunal grazing, serves to homogenize those sediments and the associated POC. Thus, we consider the estimate of 20.3 GW for aerobic respiration in the subseafloor to be applicable to sediments deeper than ~0.05-0.1 mbsf. (For comparison, our estimate of total power associated with O_2 consumption by marine sediments globally, including the bioturbated upper layer, is $3,000 \pm 800 \text{ GW}$ (see above), consistent with the notion that the calculation of Bradley et al. [70] is specific to sediments beneath the bioturbated zone.) Because the depth of the bioturbated layer can vary to depths shallower than 0.1 mbsf, based on the density and diversity of benthic fauna, a portion of the 20.3 GW attributable to aerobic respiration beneath the bioturbated layer may therefore be associated with depths shallower than 0.1 mbsf. We therefore consider it an upper limit on the contribution of aerobic respiration to sediments below 0.1 mbsf, with the true contribution likely lower. Since the O_2 respiration rate drops with depth down through the oxic zone, and since O_2 penetration >0.1 m occurs mostly in deep-sea sediments with sparse fauna, we expect that most of this “subseafloor” O_2 consumption occurs shallower than 0.1 m depth. For example, in the large ocean gyre regions only about 1% of the entire O_2 consumption takes place beneath 0.1 m [53]. Thus, we consider it likely that O_2 consumption in sediments deeper than 0.1 mbsf constitutes a small fraction of the total of 20.3 GW.

To estimate a lower bound, we consider that, in the main ocean gyre regions, the subseafloor O_2 consumption rate is so low that O_2 may be present throughout the sediment pile and down to the underlying ocean crust, such that only aerobic respiration takes place [39]. We used the estimate of D'Hondt et al. [39] that 9-37% of the abyssal seafloor may be oxic to basement to estimate the

global oxygen consumption in gyre sediments deeper than 0.1 m. The subsurface O_2 flux in ocean gyre regions ranges from ca. 2×10^{-4} to 2×10^{-2} mmol O_2 m $^{-2}$ d $^{-1}$ (data compiled by Jørgensen et al. [53]). For the present calculation, a mean value of 5×10^{-3} mmol O_2 m $^{-2}$ d $^{-1}$ or 5 mol O_2 km $^{-2}$ d $^{-1}$ was used. The maximum value of 37% of the global ocean area corresponds to 88×10^6 km 2 , which yields a global O_2 consumption beneath 0.1 meter of $(5 \times 88 \times 10^6 \times 365 =) 0.16$ Tmol O_2 yr $^{-1}$. Assuming a molar stoichiometry of 0.75 mole C oxidized per mole O_2 for aerobic carbon respiration in deep sea sediments [53] yields a carbon oxidation rate of 0.12 Tmol C yr $^{-1}$ or 3.8×10^3 mol C s $^{-1}$. Given the ΔG for aerobic respiration of marine organic carbon (-443 kJ (mol C) $^{-1}$), as described above, the power associated with aerobic respiration in these gyre sediments beneath 0.1 m is thus up to $(3.8 \times 10^3 \times 443 \times 10^3 / 10^9 =) 1.7$ GW.

Reconciling the lower limit estimate of 1.7 GW with the expectation that the true value is a relatively small fraction of the total of 20.3 GW, we adopt a value of 5 GW, about one quarter of the total aerobic power, as the power associated with aerobic respiration in sediments deeper than 0.1 mbsf. As shown below, this is a small fraction of the anaerobic power beneath 0.1 m depth, so the 3- to 4-fold uncertainty in this estimate does not drive a correspondingly large uncertainty in the total power of respiration in sediments >0.1 mbsf.

We estimate a rate of carbon oxidation with sulfate in sediments >0.1 m from the global sulfate reduction rates in the seabed, 45 Tmol SO_4^{2-} yr $^{-1}$ [79], minus the sulfate reduction taking place in the top 0.1 m, 9 Tmol SO_4^{2-} yr $^{-1}$ (estimated above). The resulting global rate is 36 Tmol SO_4^{2-} yr $^{-1}$, corresponding to 1.14×10^6 mol SO_4^{2-} s $^{-1}$. For comparison, Egger et al. [71] reported a global sulfate flux to the sulfate-methane transition depth, which is typically much deeper than 0.1 mbsf, of 5.3 Tmol SO_4^{2-} yr $^{-1}$. Analysis of their database shows that this corresponds to the global sulfate reduction beneath ~ 1 m sediment depth, and the difference between our sulfate flux estimate and that of Egger et al. [71] is consistent with the fact that our estimate includes sulfate reduction that occurs in the 0.1-1 mbsf interval while that of Egger et al. does not. We used a molar stoichiometry for organic carbon oxidation by sulfate reduction of 1.7 C/ SO_4^{2-} [74] to calculate a global organic carbon oxidation by sulfate reduction beneath 0.1 m of 61 Tmol C yr $^{-1}$, or 1.94×10^6 mol C s $^{-1}$. Given the ΔG for organic carbon oxidation by sulfate of 32.2 kJ (mol C) $^{-1}$ (see above), the total power for carbon oxidation with sulfate as the ultimate terminal electron acceptor beneath 0.1 m is $(1.94 \times 10^6 \times 32.2 \times 10^3 / 10^9 =) 62$ GW. Globally, the combined power associated with carbon oxidation with O_2 and sulfate as the ultimate electron acceptors in sediments >0.1 m is thus $(5 + 62 =) 67$ GW, of which sulfate reduction accounts for $\sim 93\%$. For comparison, Bradley et al. [70] estimated a lower global power of 37 GW for subseafloor sediments.

Additional power is produced by water radiolysis. High-energy γ and α radiation from the nuclides ^{40}K , ^{232}Th and ^{238}U occurring naturally in sediments and rocks causes radiolytic splitting of water and production of H_2 and reactive oxygen species. Based on laboratory experiments with radiation of seawater and different types of sediments, Sauvage et al. [83] made a global extrapolation of the radiolysis rate in the seabed of 2.7×10^{13} mol electron equivalents per year, which could potentially reduce 1.07×10^5 mol SO_4^{2-} s $^{-1}$. If this turnover of electrons were used in the subseafloor by sulfate reducing bacteria, it would provide an additional energy flux of 3 GW, thereby summing up to 70 GW. However, the fate of the reactive oxygen species produced is not known. They may catalyze chemical reactions that compete with a biological utilization of radiolytically produced H_2 .

Terrestrial heterotrophic respiration. Our estimates of terrestrial heterotrophic respiration include soils, humanity, and livestock but do not include contributions from wild animals (e.g., birds,

mammals, and some arthropods) that are not soil-associated. We could not identify a basis in the literature for directly estimating global rates of heterotrophic respiration by such groups; however, their contribution to the global energy budget is likely to be small. For example, Bar-on et al. [31] estimated a total mass for wild birds and mammals of 0.009 Pg C, less than 0.05% of the total biomass of terrestrial heterotrophic consumers and about 4% of the combined mass of humanity and livestock. From data in Fig 2A, mammals have a mean field MSP of $0.05 \text{ W (g C)}^{-1}$ and birds a mean field MSP of 0.2 W (g C)^{-1} . Combining these with the global mass estimates of Bar-on et al. [31] (wild mammals: 0.007 Pg C, wild birds: 0.002 Pg C) suggests a combined heterotrophic respiration rate for these groups of $\sim 0.75 \text{ TW}$, less than 1.5% of total terrestrial heterotrophic respiration.

For soils, the bulk of CO_2 fluxes emitted to the atmosphere from respiration comes from the upper 30 cm of soil, while organic matter dynamics below this depth are less studied [84]. Presently available data support an ability to constrain rates of organic carbon respiration in terrestrial soils mostly in this upper layer, but not to resolve soil respiration into specific horizons (e.g., above and below 0.1m, as in marine sediments), nor to estimate rates of heterotrophic respiration by the terrestrial “deep biosphere”. Despite potentially hosting a microbial biomass on par with that of terrestrial soils [85], the terrestrial deep biosphere likely accounts for only a very small fraction of terrestrial heterotrophic respiration. For example, if these microbes operate with an average MSP comparable to those in marine sediments $> 0.1 \text{ mbsf}$ ($0.00002 \text{ W gC}^{-1}$), it would imply a total energy utilization rate of only $\sim 1\%$ of that in soils.

The above estimates of power for the terrestrial deep biosphere and for wild animals not associated with soils are not included in Table 1 because they derive from estimates of MSP, whereas all other values for power in Table 1 are calculated independently from published estimates of global carbon fluxes or process rates. We include these calculations here simply to indicate that they likely represent only minor terms in the power budget of the global biosphere.

Terrestrial heterotrophic respiration, soils (0-8m). For heterotrophic respiration in terrestrial soils, we use the estimate of Ciais et al. [52], which is computed as the residual (difference) between NPP, Net Ecosystem Exchange, and several other regional carbon fluxes. That study reports a value of 39 Pg C yr^{-1} with an interquartile range (IQR) of $33\text{--}46 \text{ Pg C yr}^{-1}$. Approximating the standard deviation as $\text{SD} \approx \text{IQR}/1.35$ (which assumes a normal distribution) gives $\text{SD} \approx \pm 9.6 \text{ Pg C yr}^{-1}$. This converts to a heterotrophic respiration, $R_H = (1.03 \pm 0.26) \times 10^8 \text{ mol C s}^{-1}$ and a corresponding power of $50 \pm 13 \text{ TW}$. For comparison, several studies have sought to directly estimate rates of heterotrophic soil carbon respiration. Using a “top-down” remote sensing-based approach, Konings et al. [86] report a similar estimate of $43.6 \pm 19.3 \text{ Pg C yr}^{-1}$. Higher estimates are made in two studies based on up-scaling of soil CO_2 efflux measurements, with partitioning of that efflux into components contributed by heterotrophic respiration (the quantity sought here) and autotrophic respiration. This approach yields estimates of $51.5 \pm 0.6 \text{ Pg C yr}^{-1}$ [87] and $57.2 \pm 0.6 \text{ Pg C yr}^{-1}$ [88].

Livestock and humanity. The United Nations Food and Agriculture Organization (FAO) provides the data needed to estimate respiration rates by both humanity and livestock. For humanity, FAO (fao.org/faostat) gives a 2017 global population of 7.6311×10^9 and per-capita average caloric intake of 2908 kcal d^{-1} ($= 140.7 \text{ W}$). Combining the two yields a direct estimate of power (rather than carbon respiration rate) of 1.074 TW . For technological energy consumption, British Petroleum Corp. [89] reports annual worldwide and US energy use of 583.90 and 94.65 Exajoules

(18.515 TW and 3.001 TW), respectively, for calendar year 2019 (the last year without pandemic-related impacts on energy consumption). For livestock, the FAO Global Livestock Environmental Assessment Model (GLEAM; [90]) reports a global feed consumption rate of 6.0 Gt dry matter. We assume an uncertainty of ± 0.5 in this value based on the reported significant figures. Assuming a dry mass to carbon mass ratio of 2 [31] yields a carbon respiration rate of 3.0 ± 0.25 Pg C yr⁻¹ ($= (7.9 \pm 0.6) \times 10^6$ mol C s⁻¹), which yields 3.9 TW based on a conversion rate of $\Delta G = -487,000$ J (mol C)⁻¹.

3.3. Geochemical energy fluxes in crustal systems

Chemolithoautotrophic microorganisms oxidize reduced chemical species from the environment with oxygen or other oxidants and conserve the liberated energy from the process in the biological energy currency, ATP. Similar to the phototrophs, they use the conserved energy to assimilate CO₂ from the environment and produce biomass for growth. Sources of geochemical energy that can support chemolithoautotrophy arise from differences in the oxidation state between the reduced rocks that compose most of the interior of the Earth and the relatively oxidized conditions at the planet's surface. There is, to our knowledge, not a comprehensive synthesis available of the global flux of electron donors or of the potential power for the biosphere associated with their oxidation. We have therefore provided a more detailed discussion of this in the following.

The energy transfer from rocks to surface (or near-surface) environments, where the energy can be utilized by microorganisms, primarily takes the form of electron donors that are predominantly created through fluid-rock interactions. The most abundant electron donors include hydrogen sulfide (H₂S), molecular hydrogen (H₂), methane (CH₄) and dissolved ferrous iron (Fe²⁺). In surface and shallow subsurface environments, chemolithoautotrophs can combine these electron donors with electron acceptors (e.g., O₂, NO₃⁻, SO₄²⁻, CO₂) to generate metabolic energy.

It follows that quantifying the flux of electron donors constrains the amount of chemical energy that is potentially available to chemolithoautotrophs in geological environments. A compilation of estimates for the major fluxes of electron donors produced by geological processes is listed in Table S4, with descriptions of the basis for the estimates in different environments provided in the following sections. Because the geological processes that generate electron donors differ substantially between the ocean basins and the continents, sources are considered separately for these two domains.

The compilation includes only those electron donors that are produced solely by geological processes. Many chemolithoautotrophs utilize electron donors produced by other microorganisms during degradation of organic matter (e.g., H₂ generated by fermenting bacteria, or CH₄ produced by methanogenic archaea) or by thermal degradation of biologically derived organic matter. However, because these sources can be considered as part of the biosphere supported by photosynthetic primary production of biomass, they have not been included here.

In addition, this compilation focuses on those geological sources of electron donors that appear to be the largest contributors on a global scale. A number of additional geological processes that generate electron donors have been identified in the literature over the years, but these appear to be less widespread and significantly smaller in scale (e.g., H₂ production during rock comminution at the base of glaciers; [91]). Though minor globally, these sources may nevertheless support significant primary production in the local environments where they occur.

Table S4. Summary of major geochemical fluxes of electron donors that can potentially support chemolithoautotrophic metabolism.

	Chemical Flux (10^{11} moles yr^{-1})			
	H ₂	CH ₄	H ₂ S	Fe(II)
<i>Oceanic Realm</i>				
Hydrothermal (basalt-hosted)	0.43	0.11	4.3	1.8
Hydrothermal (ultramafic-hosted)	0.58	0.06	0.18	0.4
Basaltic crust alteration	4.5	?	1.7	8.5
Serpentinization (ocean basins)	3.4	0.37	?	?
Serpentinization (subduction zones)	0.8	0.3	?	?
Radiolysis	0.005	0	0	0
<i>Continental Realm</i>				
Deep-subsurface hydration (incl.serpentinization)	0.2	0.2	?	?
Ophiolites	0.17	0.015	?	?
Radiolysis	0.3	0	0	0
Volcanic/geothermal	0.8	0.3	0.5	?
Shallow aquifers in crystalline rocks	0.6	?	0.15	1.2

“?” designates electron donors that may be present but for which no reliable information on global annual fluxes could be found. However, the magnitude of these fluxes is expected to be small relative to other quantified fluxes listed in the table.

The fluxes of electron donors were converted to energy potentially available to support chemolithoautotrophy by multiplying the fluxes by the Gibbs energy of a representative metabolic reaction. Results of the calculations are summarized by electron donor and geologic environment in Table S5, with details on the calculations provided in a subsequent section. To reflect the inherent uncertainties in the electron donor fluxes, data in this table have been truncated to two significant digits, while the summary data in Main Text Table 1 have been truncated to one significant digit.

3.3.1 Oceanic domain

In the oceanic domain, the creation of new ocean crust along mid-ocean ridges generates a continuous supply of electron donors. One primary source of these electron donors is the hydrothermal systems that develop along the mid-ocean ridges. Here, the heat generated by the upwelling of hot rock and magma along the mid-ocean spreading center creates hydrothermal circulation cells that draw cold seawater into the ocean crust, where it is heated and returns to the surface. During circulation, fluid-rock interactions result in production of dissolved electron

donors. Submarine hydrothermal activity also occurs in isolated volcanic regions (hotspots) that occur across the globe (e.g., Hawaii, Réunion, Italy, etc.), but the net flux of fluids in these systems is very small compared to mid-ocean ridge systems, so they are not included here.

Table S5. Fluxes of chemical energy potentially available for chemolithoautotrophy in geological systems.

	Energy Flux (GW)				Total Flux (GW)
	H ₂	CH ₄	H ₂ S	Fe	
<i>Oceanic Realm</i>					
Hydrothermal (basalt-hosted)	0.3	0.28	10.6	0.18	11
Hydrothermal (ultramafic-hosted)	0.42	0.15	0.44	0.04	1.1
Low-temperature basaltic crust alteration	3.3	?	4.2	0.85	8.3
Serpentinization (ocean basins)	2.50	0.95	?	?	3.3
Serpentinization (subduction zones)	0.58	0.77	?	?	1.4
Ocean totals	7.1	2.2	15.2	1.1	26
<i>Continental Realm</i>					
Deep-subsurface hydration (incl. serpentinization)	0.15	0.52	?	?	0.67
Ophiolites	0.12	0.04	?	?	0.16
Radiolysis	0.22	?	?	?	0.22
Volcanic/geothermal	0.58	0.77	1.2	?	2.6
Shallow aquifers in crystalline rocks	0.44	?	0.37	0.12	0.93
Continent totals	1.5	1.3	1.6	0.12	4.5
Global totals	8.6	3.5	16.8	1.2	30

"?" designates energy sources for which no reliable information on global annual fluxes could be found, most of which are probably small relative to the other energy sources identified.

Additional electron donors are generated as the ocean crust ages and cools as it migrates away from the mid-ocean ridge. In these environments, fluid circulation continues at lower temperatures, and reacts with the surrounding rock to generate electron donors. These reactions involve low-temperature alteration of both mafic rocks (basalts and gabbros) and ultramafic rocks (mostly harzburgites and lherzolites). The latter process is referred to as serpentinization, owing to the predominance of the mineral serpentine in the reaction products.

Mid-Ocean ridge hydrothermal systems

Deep-sea hydrothermal systems along the mid-ocean ridges are a prolific source of electron donors and are recognized to support large communities of chemoautotrophic microorganisms. Deep-sea hydrothermal systems form where the spreading of Earth's tectonic plates allow hot rocks to be

exposed near the seafloor, leading to circulation of seawater into the crust. During circulation, the reaction of seawater with rocks produces electron donors. Electron donors that occur in high abundance in the hydrothermal fluids include H_2 , H_2S , CH_4 , and Fe^{2+} .

The rocks that host seafloor hydrothermal systems occur in two main types: mafic (basalt, gabbro) and ultramafic (peridotites). The abundance of electron donors in hydrothermal fluids differs between the different host rocks (mafic rocks typically have higher H_2S and ultramafic rocks have higher H_2 and CH_4 ; Table S6), so they are generally treated separately when estimating fluxes. Submarine hydrothermal systems also occur in other settings, such as the submerged flanks of volcanoes at hotspots (e.g., Hawaii) or subduction zones (e.g., Vulcano), but the annual flux of fluids through these systems is small compared with mid-ocean ridges, so they are not included here.

A number of studies have estimated the global fluxes of electron donors in deep-sea hydrothermal systems, focusing mainly on H_2 (e.g., [82, 92-94]). The typical approach for estimating electron donor fluxes involves determining their average composition in hydrothermal fluids and then multiplying those concentrations by the annual flux of hydrothermal fluids discharged globally. The average composition of electron donors is relatively well constrained because numerous seafloor hydrothermal vents have been sampled over the years; a compilation of currently available data is provided in [Supplementary Dataset S2](#). Estimates of net fluid fluxes have largely been based on the heat loss that occurs along mid-ocean ridges from cooling of igneous rocks and loss of residual heat from the mantle ([95-96], and references therein). Substantial uncertainties in the net flux arise from limited knowledge of these parameters, particularly the mechanisms for heat transfer from the lower ocean crust and the balance of relative contributions from different heat sources in different settings. Uncertainties also arise from estimates of the fractions that discharge through mafic versus ultramafic rocks.

Here, we adopt a total hydrothermal fluid flux of $5.6 \times 10^{13} \text{ kg yr}^{-1}$ from German and Seyfried [96]. Of this total, only a relatively small fraction ($\sim 20\%$) is discharged at high-temperature ($>250^\circ\text{C}$) sites of focused flow (“black smokers”), while the remainder is discharged at lower temperatures at “diffuse” vents (mostly $5\text{-}100^\circ\text{C}$). The diffuse vents mostly represent higher temperature fluids that have cooled and/or mixed with seawater in the shallow subsurface prior to venting. Although the composition of diffuse fluids can be influenced by biological consumption and abiotic reactions along the flow path, this has a relatively minor impact on the net amount of electron acceptors made available at the ridge axis. We also adopt an estimate from Keir [92] that 92% of the total flux discharges from mafic-dominated systems and 8% from ultramafic systems.

Average concentrations of electron donors in hydrothermal fluids in mafic- and ultramafic-hosted systems are shown in Table S6. Combining these compositions with the fluid fluxes yields the estimated electron fluxes from mid-ocean ridges shown in Table S4. The values calculated here are similar to earlier estimates of Canfield et al. [82] and Keir [92]. However, the H_2 flux estimate is about an order of magnitude lower than that of Worman et al. [94], who proposed a total on-axis hydrothermal flux of $9.4 \times 10^{11} \text{ moles H}_2 \text{ yr}^{-1}$. The principal factor accounting for this difference is that Worman et al. [94] adopt a hydrothermal fluid flux through on-axis systems that is about an order of magnitude larger than that used here and in other studies. This higher flux estimate encompasses both higher temperature focused flow and lower temperature diffuse discharge. Worman et al.’s calculation assumes the same H_2 concentrations for all discharged fluids, even though diffuse vents typically have substantially lower H_2 concentrations (e.g., [97]) consequently, their on-axis flux estimates appear to be too high.

Some of the Fe and H₂S discharged by deep-sea hydrothermal vents precipitates out of solution as metal sulfide minerals when the hydrothermal fluids mix with seawater. However, this does not necessarily represent a significant reduction in the amount of chemical energy made available to microorganisms since the mineral precipitates can be colonized by autotrophic organisms that oxidize them to gain metabolic energy (e.g., [98]).

Table S6. Average concentrations of electron donors in endmember mid-ocean ridge hydrothermal fluids (compiled from data in [Supplementary Dataset S2](#)).

	Concentration (mmol kg ⁻¹)			
	H ₂	CH ₄	H ₂ S	Fe
Basalt-hosted systems	0.83	0.22	8.4	3.6
Ultramafic-hosted systems	14	1.5	4.0	9.0

Low-temperature basaltic crust alteration

Fluids continue to circulate through the oceanic lithosphere as it ages and migrates away from the mid-ocean ridge, driven by residual heat in the crust and underlying mantle. The exact nature of this circulation (flow paths, fluxes, residence times, etc.) remains largely unknown. Lower temperature fluid-rock interactions continue in this realm and produce additional electron acceptors, primarily H₂S, H₂ and Fe²⁺ [99]. Seawater-derived fluids circulating through the aging oceanic lithosphere in this realm also transport electron acceptors into the crust that organisms can combine with electron donors to gain metabolic energy.

Bach and Edwards [99] have estimated fluxes of electron acceptors provided by low temperature alteration of basaltic rocks in this realm, and their estimates are adopted here. Based on examination of subseafloor rocks recovered by drilling, these authors estimate that $1.7 \pm 1.2 \times 10^{12}$ moles Fe(II) and $1.1 \pm 0.7 \times 10^{11}$ moles S from oceanic basalts are oxidized each year, which is utilized to at least some degree by microorganisms. Assuming that 50% of the reducing power in Fe(II) is converted to H₂ by hydrolysis, Bach and Edwards [99] estimate that $4.5 \pm 3.0 \times 10^{11}$ moles H₂ are generated within the aging ocean crust per year. Because the estimates of Fe and S oxidation are constrained by numerous drill core samples, the estimates probably have relatively low uncertainty (a factor of 2-3 or so), but the fraction converted to H₂ is largely unconstrained.

Serpentinization

In addition to mafic rocks (basalts and gabbros), ocean lithosphere also includes ultramafic rocks (i.e., peridotites) that represent remnants of the mantle that have been uplifted to the shallow subsurface by tectonic processes associated with seafloor spreading. At temperatures below 350-400°C, these rocks become unstable in the presence of water, and undergo alteration by a process known as serpentinization. Electron donors, predominantly H₂ and CH₄, are generated as a by-product of the serpentinization reaction (see [100], and references therein).

Some serpentinization takes place at high temperatures during hydrothermal circulation near the mid-ocean ridge axis, where the electron donors are discharged to the seafloor in ultramafic-hosted

hydrothermal vent fluids (Section 3.1.1). However, serpentinization continues at lower temperatures in the ocean basins as the lithosphere ages and seawater continues to circulate through the rocks. In some cases, the resulting fluids discharge to the seafloor at warm, off-axis hydrothermal vents like those at Lost City ($\sim 100^\circ\text{C}$; [101]), but elsewhere they may be discharged in diffuse vents, remain or be consumed within the lithosphere, or simply percolate diffusively out of the crust over large areas (e.g., [102]). Substantial serpentinization also occurs in subduction zones, both within the ocean lithosphere owing to fractures created by bending of the plate and in the mantle wedge that overlies the subduction zone.

Because electron donors produced by serpentinization are widely recognized to support chemolithoautotrophic communities at deep-sea hydrothermal vents and elsewhere, several studies have estimated the annual flux of H_2 generated by serpentinization of ultramafic rocks in the ocean crust (e.g., [82, 93, 94, 103]). Estimates from these studies vary somewhat but are within an order of magnitude of one another, ranging from $0.8\text{--}1.3 \times 10^{11}$ moles H_2 yr^{-1} [82] to 1.9×10^{11} moles H_2 yr^{-1} [103] to 10×10^{11} moles H_2 yr^{-1} [93–94]. Emmanuel and Ague [104] estimated a net annual production of CH_4 from serpentinization of 0.85×10^{11} moles yr^{-1} . In these studies, the fraction of electron donors that is discharged from hydrothermal vents along the mid-ocean ridge axis is not clearly differentiated from that generated elsewhere in the ocean lithosphere. In addition, while it is increasingly recognized that other electron donors such as formate and CO support microbial activity in some serpentinizing systems [105–106], there is little data on their concentrations in serpentinite-hosted fluids and there are currently no estimates for the annual fluxes of these compounds.

In general, the H_2 flux estimates are based on multiplying the volume of rock that has undergone serpentinization by the amount of H_2 generated from a given amount of serpentinized rock. Substantial uncertainties involved in the calculations include the volume of ocean lithosphere that is composed of ultramafic rocks, the fraction of these rocks that undergo serpentinization, and the amount of H_2 produced from each increment of peridotite undergoing serpentinization. With respect to the latter, most studies have assumed a uniform reaction stoichiometry for serpentinization that results in relatively high yields of H_2 per unit rock, even though the amount of H_2 generated from a given volume of rock during serpentinization is known to vary substantially as a function of a number of factors including temperature, rock composition, and pH (e.g., [107]). Much of the serpentinization that occurs in the ocean lithosphere away from the ridge axis proceeds at temperatures below $\sim 200^\circ\text{C}$, where the amount of H_2 produced from each increment of rock is significantly diminished [107–108]. Such factors have not typically been taken into account in flux estimates, which may result in published H_2 fluxes being overestimated by a factor of two or more.

Here, we adopt a net production rate from serpentinization at the lower end of the range of published studies (4×10^{11} moles H_2 yr^{-1}). From this net flux, we subtract the amount of H_2 accounted for by ultramafic-hosted hydrothermal vents, which represents serpentinization of ultramafic rocks at high temperatures along the ridge axis. The resulting estimate for generation from serpentinization in the ocean basins is 3.4×10^{11} moles H_2 yr^{-1} . Assuming a H_2/CH_4 ratio similar to that of ultramafic-hosted high-temperature hydrothermal vents, we further estimate a CH_4 flux of 0.37×10^{11} moles yr^{-1} .

Merdith et al. [109] used tectonic reconstructions of the ocean basins to estimate H_2 fluxes over the last 200 million years (Ma). Their estimates for present-day annual fluxes of H_2 (7×10^{11} moles H_2 yr^{-1}) and CH_4 ($0.7\text{--}1.2 \times 10^{11}$ moles yr^{-1}) are comparable to other studies. However, their results suggest that global H_2 fluxes from ocean crust serpentinization have probably been highly variable

throughout geologic history, ranging between 1 and 70×10^{11} moles yr^{-1} over the past 200 Ma. Presumably, CH_4 fluxes would have been similarly variable.

Less effort has been made to date to estimate fluxes of electron donors from serpentinization associated with subduction of ocean crust. However, Brovarone et al. [110] provided preliminary flux estimates of $0.045\text{--}1.5 \times 10^{11}$ moles $\text{H}_2 \text{ yr}^{-1}$ and $0.014\text{--}0.63 \times 10^{11}$ moles $\text{CH}_4 \text{ yr}^{-1}$. Here, we provisionally adopt intermediate values within these ranges (0.8×10^{11} moles $\text{H}_2 \text{ yr}^{-1}$ and 0.3×10^{11} moles $\text{CH}_4 \text{ yr}^{-1}$), with the understanding that future efforts may result in estimates that may deviate by an order of magnitude or more from these values as more information becomes available.

Radiolysis

Similar to marine sediments, radiolysis in the crystalline ocean crust can produce H_2 to support microbial activity. However, the potential for radiolytic H_2 production is substantially less in the crystalline ocean crust owing to limited pore space and lower concentrations of radioactive elements. Dzaugis et al. [111] estimated volumetric electron production rates from radiolysis in the basaltic ocean crust to be $10^{-7}\text{--}10^{-10}$ moles $\text{e}^- \text{ m}^{-3} \text{ yr}^{-1}$ for the estimated fracture widths (see their Figure 5). This is several orders of magnitude lower than the e^- flux estimated for oxidation of Fe in subseafloor basalts [99]. The range in estimates for volumetric electron production rates reflect uncertainties in the concentration of radioactive elements, extent of alteration (which increases the abundance of ^{238}U and ^{40}K), and the amount and spatial distribution of water-filled pore spaces, leading to uncertainties of several orders of magnitude for production rates using a reasonable range of values for these parameters [111].

Based on these electron production rates and a volume of basaltic oceanic crust of $1.7 \times 10^8 \text{ km}^3$ (from [94]), the estimated total annual H_2 production from crustal marine radiolysis is between 8.5×10^6 and 8.5×10^9 moles $\text{H}_2 \text{ yr}^{-1}$, with an intermediate estimate adopted here of 4×10^8 moles $\text{H}_2 \text{ yr}^{-1}$. This calculation assumes that the flux of electrons from radiolysis does not diminish over time as the crust ages, which may or may not be valid. The estimated radiolytic H_2 flux from basaltic ocean crust is about five orders of magnitude lower than that estimated for marine sediments (1.35×10^{13} moles $\text{H}_2 \text{ yr}^{-1}$; [83]), largely owing to significantly less pore space and lower K and U abundances. In addition, the estimate for radiolytic production in marine sediments includes parameters that account for enhancement of H_2 production by mineral catalysts, which is not considered in the volumetric electron flux estimates of Dzaugis et al. [111]. If minerals within the basalt are capable of catalyzing H_2 production, the volumetric H_2 fluxes may be greater than the estimates reported by Dzaugis et al. [111].

The annual flux of H_2 from radiolysis in basaltic ocean crust calculated here is several orders of magnitude lower than the estimate by Worman et al. [94] of 5×10^{11} moles $\text{H}_2 \text{ yr}^{-1}$. Worman et al. [94] also base their estimated flux on the volumetric electron production rates provided by Dzaugis et al. [111]. However, Worman et al. assume that the units in Figure 5 of Dzaugis et al. are incorrect (i.e., they infer that the units should be moles $\text{e}^- \text{ cm}^{-3} \text{ y}^{-1}$ rather than moles $\text{e}^- \text{ m}^{-3} \text{ yr}^{-1}$ as labelled). Their argument for incorrect units is based on a presumed mismatch with the Fe oxidation results shown in the same figure and those they calculate using information given by Bach and Edwards [99]. However, as shown in the “Notes” worksheet of Supplementary Dataset S2, the units in Figure 5 of Dzaugis et al. are actually correct and consistent with Bach and Edwards [99]. The incorrect inference by Worman et al. [94] leads them to overestimate the H_2 flux rate from radiolysis by a factor of 10^6 (i.e., the difference between cm^3 and m^3). The lower H_2 flux calculated here is in line with the conclusion of Dzaugis et al. [111] (illustrated in their Figure 5) that the net

contribution of electron donors from radiolysis is far smaller than that from low-temperature ocean basalt alteration.

Additional sources

Several other sources of electron donors have been identified within the ocean domain that could potentially support chemolithoautotrophic microbial activity, including rock fracturing and conversion of pyrrhotite to pyrite at hydrothermal vents. However, the net flux of these electron donors appears to be very small compared to the larger sources discussed above (e.g., [94]), and so are not considered further here.

Worman et al. [94] identified crystallization of mafic rocks (basalts and gabbros) as the largest source of H_2 within the ocean crust, which they estimate to provide a net flux of 30×10^{11} moles H_2 yr^{-1} . If valid, this source would exceed all other sources listed in Table S4 combined. Furthermore, it would greatly exceed all known fluxes of H_2 discharged from the ocean crust at the seafloor from hydrothermal vents and elsewhere. The estimate from Worman et al. [94] is based on an earlier study of Holloway and O'Day [112], who noted that the Fe(III):Fe_{total} ratio in the glassy rims of basalts erupted onto the seafloor increased during emplacement. Holloway and O'Day [112] attributed this increase to reaction of Fe(II) in basalt with seawater, resulting in generation of H_2 ($2FeO + H_2O \Rightarrow Fe_2O_3 + H_2$). Holloway and O'Day [112] estimated that this process generates ~ 300 moles H_2/m^3 of altered basalt.

Worman et al. [94] infer that similar processes occur during crystallization of the entire igneous component of the ocean crust, including the sheeted dikes and gabbroic rocks that extend several kilometers beneath the seafloor and are by far the largest volumetric components of the ocean crust. However, there is no indication that there is any significant interaction of seawater with crystallizing magma in the subsurface like that observed on the rims of basalt erupted onto the seafloor. Neither is there any evidence for significant increase in the Fe(III):Fe_{total} ratio caused by reaction with water in sheeted dikes or in the gabbroic rocks that crystallize from magma in the subsurface. Although oxide-rich gabbroic rocks with relatively high abundances of magnetite are a significant component of the lower oceanic crust, the relatively high Fe(III):Fe_{total} ratios in these rocks are predominantly attributable to removal of Fe(II) by precipitation of olivine and pyroxene during fractional crystallization, leaving the residual magma enriched in Fe(III). There is no indication that reaction with water is a significant contributor to Fe(III) production in gabbroic rocks in the lower ocean crust.

Given the absence of any clear supporting evidence for generation of significant amounts of H_2 during crystallization for most of the igneous ocean crust, we have not included this as a potential electron donor source in this study. However, should future studies confirm the large H_2 flux from crystallization proposed by Worman et al. [94], it could increase the estimated net flux of electron donors available to support chemolithoautotrophy near the seafloor by a factor of two or more.

3.3.2 Continental realm

Electron donors are also generated from geologic processes in continental settings, although the nature of fluid-rock interactions and environments for utilization of the resulting chemical energy sources are significantly different from those in the oceanic realm.

Hydration/Serpentinization

As in the ocean crust, circulation of aqueous fluids through crystalline rocks in continental crust results in the production of electron donors. Studies over the last couple of decades have documented the occurrence of deeply circulating fluids with long residence times deep within the cratons that form the cores of the continents ([113], and references therein). In many cases, these fluids contain substantial concentrations of abiotically generated H_2 . Because the cratons include ultramafic rocks, serpentinization is thought to be the source of a large fraction of this H_2 . However, H_2 can also be generated during interaction of fluids with Fe(II)-bearing minerals in other types of rock that comprise a substantial fraction of the cratons, including mafic and granitic rocks.

Sherwood Lollar et al. [113] estimated the annual production of H_2 from fluid-rock interactions within the cratons to be $0.2\text{--}1.8 \times 10^{11}$ moles H_2 yr^{-1} , noting that the magnitude of H_2 generation in this environment is comparable to that supplied by serpentinization of ultramafic rocks in the ocean basins (but see [93], for an alternative view). Here, we adopt an intermediate value from their estimated range (1.0×10^{11} moles H_2 yr^{-1}).

In addition to H_2 , fluids found in ancient cratonic rocks contain substantial amounts of CH_4 , which in many cases is present in concentrations an order of magnitude or more higher than those of H_2 (e.g., [113]). Presumably, this CH_4 is largely derived from reduction of dissolved CO_2 by the H_2 generated through water-rock interactions. Methanogenic microorganisms may be involved in the formation of some of this CH_4 , but there is also evidence for contributions from abiotic synthesis (e.g., [113]). At present, it is uncertain how much of the H_2 produced by fluid-rock interactions within cratons is converted to CH_4 , or how much of this conversion is attributable to microbial versus abiotic processes. In the absence of any firm constraints, we make the simple assumption that 80% of the H_2 is converted to CH_4 , which is then available as a potential electron donor for microbial metabolism (note that four moles of H_2 are required to convert one mole of CO_2 to CH_4). The amount of energy available from aerobic oxidation of CH_4 on a per-electron basis is similar to that from H_2 (see below), so this assumption has only a very small impact on the estimate for net amount of chemical energy available from cratonic rocks. Fluid-rock interactions by deeply circulating fluids within cratons may also make reduced Fe and S available for exploitation by chemolithoautotrophs, but estimates of the amounts of these components are currently unavailable.

Ophiolites

In addition to the cratons, ultramafic and mafic rocks that generate electron donors through serpentinization and other mineral hydration reactions occur in subaerial setting in ophiolites, which represent sections of the ocean crust that have been obducted onto continents. For the most part, ophiolites occur on the margins of continents or on islands such as New Zealand. Many ophiolites have already experienced some degree of serpentinization in the ocean basins prior to obduction, but enough fresh rock remains for substantial serpentinization to continue after obduction. The extent of pre-obduction serpentinization and the rate of reaction after obduction remain areas of active research.

Groundwater and springs discharged from ophiolites commonly contain elevated concentrations of H_2 and abiotic CH_4 produced by serpentinization and other fluid-rock interactions, some of which may reflect residual volatiles generated during serpentinization in the ocean basins prior to obduction. While H_2 and CH_4 fluxes have been measured for a few local systems, estimates of the global annual fluxes of these compounds from ophiolites are not currently available ([114]). However, given that the total volume of ophiolites is small relative to ocean crust, annual fluxes of electron donors are likely to be substantially smaller than other sources considered in this study. As

a provisional estimate, we assume here a flux of H_2 and CH_4 from ophiolites equal to 5% of that of the ocean basins.

Radiolysis

As for seafloor sediments and ocean crust, radiolysis is a potential source of electron donors in continental settings, particularly for H_2 [115]. Sherwood Lollar et al. [113] estimated the annual flux of H_2 generated by radiolysis in cratons to be $0.16\text{--}0.47 \times 10^{11}$ moles $\text{H}_2 \text{ yr}^{-1}$. This estimate does not include production in sedimentary basins on land, which commonly have relatively high concentrations of radioactive elements (^{40}K , ^{238}U , ^{232}Th , etc.). We adopt here an intermediate value of 0.3×10^{11} moles $\text{H}_2 \text{ yr}^{-1}$ for radiolysis in the continental domain, with the understanding that production in continental sedimentary basins may increase this value. The flux of H_2 from radiolysis in cratons is substantially higher than that in the ocean crust owing to the higher abundance of radioactive elements and the larger volume, but much smaller than the flux from marine sediments.

In addition to the cratonic cores and sedimentary basins of the continents, radiolysis may generate H_2 in other types of rocks that make up the continents, including orogenic belts and volcanic provinces. To date, however, there have evidently been no estimates of the potential for radiolytic H_2 production in these provinces, and it would be difficult to produce reasonable provisional estimates based on available data. Consequently, this electron donor source is not included here, but it could represent a significant additional source of H_2 in continental settings beyond the cratons.

Volcanoes and geothermal systems

Volcanoes and geothermal systems on land discharge significant amounts of electron donors, including H_2 , CH_4 , and H_2S , occurring both as components of gaseous vapors and as dissolved species in hydrothermal fluids. Abiogenic sources of these compounds include volatiles released during crystallization of magma and fluid-rock interactions that occur during hydrothermal circulation of groundwater induced by the heat released by magmatic intrusions. In many cases, however, a substantial fraction of the electron donors in these systems, particularly CH_4 and other hydrocarbons, may derive from biogenic sources or from thermal decomposition of organic matter (thermogenesis) that is either transported by circulating fluids or resides in rocks intruded by magmas and hydrothermal fluids ([116], and references therein).

Canfield et al. [82] estimated annual fluxes of $0.9\text{--}3.4 \times 10^{11}$ moles $\text{H}_2 \text{ yr}^{-1}$ and $0.4\text{--}0.6 \times 10^{11}$ moles $\text{H}_2\text{S} \text{ yr}^{-1}$ from subaerial volcanoes based on the ratio of these components relative to SO_2 in volcanic emissions together with the annual flux of SO_2 from volcanoes. Based on a comprehensive database of measured CH_4 abundances in volcanic and geothermal systems, Etiope et al. [117] estimated an annual abiotic flux of 0.25×10^{11} moles $\text{CH}_4 \text{ yr}^{-1}$. This value takes into account the substantial fraction of CH_4 that is evidently derived from biogenic and thermogenic sources in these systems. The values adopted here are based on these estimates. It should be acknowledged that a substantial fraction of the volatile electron donors generated in volcanic environments pass rapidly through the system and enter the atmosphere. Nevertheless, these gaseous compounds may still be available to some degree to organisms that are capable of extracting metabolic energy from H_2 and CH_4 in the atmosphere (e.g., [118], and references therein).

Shallow crystalline rocks

Outside of the cratons that are the core of continental land masses, there are other crystalline rocks (i.e., non-sedimentary rocks) that can produce electron donors to support autotrophic microbial communities. One well-studied sample is the layered basalts of the Snake River Plain in the northwest U.S., where reaction of basaltic rocks with groundwater is thought to support indigenous populations of methanogens and other chemolithoautotrophs supported by H_2 produced by basalt-water interactions (e.g., [119-120]). Large provinces of basaltic rocks occur in places like the Snake River and Deccan Traps, with many smaller occurrences scattered around the globe. In addition to basalts, fluid-rock interactions can generate H_2 and other electron donors from other rock types including rhyolites and granites. Large-scale occurrences of these types of rock occur in mountain belts above subduction zones (e.g., the Andes, Coast Range in western North America) and in orogenic belts. Note that we refer here to the production of electron donors separate from that which occurs through volcanic and geothermal activity when these rocks were originally emplaced, which is accounted for elsewhere. Here, we refer to the continued reaction of the rocks with groundwater after the initial heat from crystallization has dissipated.

At present, there does not appear to be any published estimates of the global production of electron donors from low-temperature fluid-rock interaction in continental crystalline rocks outside of that occurring in the cratonic cores. For this study, we provisionally estimate electron fluxes from these environments based on analogy with weathering of seafloor basalts [99]. Average annual rates of production for igneous rocks in subaerial settings have been estimated by White et al. [121] to be $0.9 \text{ km}^3 \text{ yr}^{-1}$ for flood basalts, $0.026 \text{ km}^3 \text{ yr}^{-1}$ for other basalts, $0.0023 \text{ km}^3 \text{ yr}^{-1}$ for andesites, and $0.0004 \text{ km}^3 \text{ yr}^{-1}$ for granites. Since andesites and granites make up a small fraction of the total and have significantly lower concentrations of Fe and S to generate electron donors, we neglect these components in the flux estimate here. Additionally, owing to low porosity and permeability that limit fluid access to the interiors of basaltic rocks, we assume that only about 20% of the basalts adjacent to fractures are accessible to liberate electron donors. Following Bach and Edwards [99], we assume that ~50% of the Fe oxidation produces H_2 . Based on these assumptions, we adopt provisional estimates of average annual electron donor production from shallow crystalline rocks to be 0.6×10^{11} moles $H_2 \text{ yr}^{-1}$, 0.15×10^{11} moles $H_2S \text{ yr}^{-1}$, and 1.2×10^{11} moles $Fe(II) \text{ yr}^{-1}$.

3.3.3 Energy fluxes in crustal systems

The fluxes of electron donors (Table S4) were converted to the potential power supply available to support chemolithoautotrophy (Table S5) by multiplying the flux of each electron donor by the Gibbs energy of a representative metabolic reaction utilising that electron donor. Gibbs energies for the metabolic reactions were calculated using the methods and data given by Amend and Shock [122]. The energy available from the reactions can vary depending on the local chemical environment and the electron acceptor that couples with the donor (e.g., [122]).

The Gibbs energy (ΔG) of potential metabolic reactions can be calculated using the formula:

$$\Delta G = \Delta G^0 + RT \ln Q$$

where ΔG^0 is the standard Gibbs energy and Q is the quotient of the activities of products and reactants involved in the reaction [123]. In natural environments, local concentrations of electron donors, electron acceptors, and other species involved in metabolic reactions can be highly variable. For this study, it was assumed that all electron donors were consumed using O_2 under locally microoxic

conditions ($100 \mu\text{mol O}_2 \text{ kg}^{-1}$) at 25°C and $\text{pH} = 7$. It is also assumed that each electron donor has a local concentration of $100 \mu\text{mol kg}^{-1}$. Concentrations of product species are assumed to be $\text{HCO}_3^- = 1 \text{ mmol kg}^{-1}$, $\text{SO}_4^{2-} = 28 \text{ mmol kg}^{-1}$, and $\text{Fe}^{3+} = 1 \text{ nmol kg}^{-1}$. For simplicity, an activity coefficient equal to one is assumed for all aqueous species.

The metabolic reactions and their calculated Gibbs energies are summarized in Table S7. Because the Gibbs energy of the reactions is relatively insensitive to the concentrations of reactants and products over a range of several orders of magnitude (Fig. S5), the choice of concentrations for the calculations has a relatively small impact on the power flux calculations relative to other factors. However, it should be noted that significantly less energy would be available when alternative electron acceptors are used rather than O_2 (e.g., NO_3^- or SO_4^{2-}).

Table S7. Calculated Gibbs energy per mole electron donor for microbial utilization of the electron donors considered in this study.

Metabolism	Reaction	ΔG (kJ/mole)	ΔG (kJ/mole e^-)
Hydrogen oxidation	$\text{H}_{2(aq)} + 0.5\text{O}_{2(aq)} \rightarrow \text{H}_2\text{O}$	-229	-115
Methane oxidation	$\text{CH}_{4(aq)} + 2\text{O}_{2(aq)} \rightarrow \text{HCO}_3^- + \text{H}^+ + \text{H}_2\text{O}$	-811	-101
Sulfide oxidation	$\text{HS}^- + 2\text{O}_{2(aq)} \rightarrow \text{SO}_4^{2-} + \text{H}^+$	-775	-97
Iron oxidation	$\text{Fe}^{2+} + 0.25\text{O}_{2(aq)} + \text{H}^+ \rightarrow \text{Fe}^{3+} + 0.5\text{H}_2\text{O}$	-31	-31

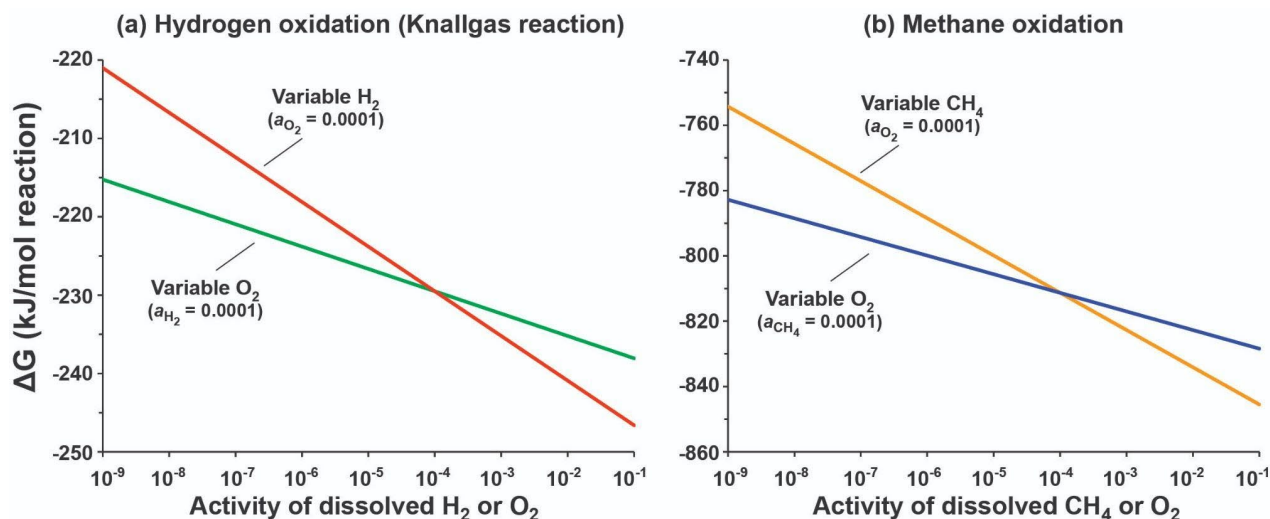


Figure S5. Gibbs energy for (a) hydrogen oxidation and (b) methane oxidation calculated for variable activity of dissolved O_2 with electron donor constant and for variable activities of the electron donor with O_2 constant. See [Supplementary Dataset S2](#) for calculations.

4. Mass-Specific Power (MSP) Calculations

4.1. Source data and MSP calculations for Figure 3

Cultures:

Aerobic glucose oxidizers

Geobacillus LC300 [16], *Vibrio natriegens* [18], *Escherichia coli* W3110 [17]: Reported values of mass-specific glucose consumption rate (as grams glucose C per gram biomass C per time) were converted to MSP using a factor of $39.6 \text{ kJ (g glucose C)}^{-1}$, equivalent to the Gibbs energy change for oxidation of 50 mM glucose at 0.2 bar O_2 .

Sulfate-reducing bacteria

Desulfobacter psychrotolerans [124]: Reported values of growth yield (as grams dry weight per mole acetate) and specific growth rate were used to calculate a mass-specific acetate consumption rate for each of the three temperatures (5, 15, and 20°C) at which both of those quantities were measured. The mass-specific acetate consumption rate was converted to MSP using a biomass conversion factor of 2 grams dry weight per gram C and an estimated Gibbs energy change of -61.7, -62.2, and -62.5 $\text{kJ (mol acetate)}^{-1}$, at 5, 15, and 20°C , respectively. The Gibbs energy change estimates are calculated at the respective temperatures based on $\Delta G^\circ = -47.6 \text{ kJ (mol acetate)}^{-1}$ [125] and culture conditions of $[\text{acetate}] = [\text{sulfate}] = 20\text{mM}$, $[\text{HCO}_3^-] = 30\text{mM}$, and $[\text{HS}^-] = 1\text{mM}$.

Desulfobacterium autotrophicum [126]: Reported values of mean cell-specific sulfate reduction rate for lactate-limited chemostat growth were converted to mean cell-specific power using a Gibbs energy change of $-78.5 \text{ kJ (mol sulfate)}^{-1}$, the midpoint in the range of -76 to $-81 \text{ kJ (mol sulfate)}^{-1}$ that was calculated by Marietou et al. [126] for the conditions within the chemostat. Conversion to MSP is based on an estimated mean cell mass of 60 fg C.

Methanogenic archaea

Methanococcus maripaludis [23]: Reported values of mass-specific methane production rate were converted to MSP using a dry mass to carbon mass conversion factor of 2 and an estimated Gibbs energy change for the chemostat conditions of $-40 \text{ kJ (mol CH}_4)^{-1}$. The Gibbs energy estimate is intermediate between the most negative value that is permissible based on the reported chemostat conditions (-56.7 to -53.9) and an assumed maximum of $-30 \text{ kJ (mol CH}_4)^{-1}$. The value of $-30 \text{ kJ (mol CH}_4)^{-1}$ based on the reported stoichiometry of ATP formation (0.5 ATP per methane; Müller et al., 2021) and the contention that a minimum of $60 \text{ kJ (mol ATP)}^{-1}$ are required for ATP synthesis in growing cells [127].

Methanobacterium thermoautotrophicum [21]: Reported values of growth yield (as grams dry weight per mole methane) and specific growth rate were used to calculate a mass-specific methane production rate. This rate was converted to MSP using a biomass conversion factor of 2 grams dry weight per gram C and an estimated Gibbs energy change of $-112.7 \text{ kJ (mol CH}_4)^{-1}$, which was calculated from the reported culture conditions at the experimental temperature of 65°C .

In all cases for culture-based studies, we plot specific growth rate (the quantity reported in the studies) in place of specific carbon turnover rate. These two quantities differ in that specific carbon turnover rate, as we define it, encompasses internal carbon turnover (e.g., breakdown and new synthesis of proteins that does not lead to a net change in carbon biomass) as well as net growth, whereas specific growth rate does not include internal carbon turnover. Therefore, specific growth rate underestimates (and places a lower limit on) the true specific carbon turnover rate. However, we consider it a close approximation in each of the cases plotted in Fig. 4. For example, in the case of the lowest methanogen culture MSP in Fig. 4 (for *Methanococcus maripaludis* growing at 1% of its maximum rate), which has the second lowest MSP among all cultures plotted on Fig. 4, Müller et al. [23] report that net growth still accounted for >70% of total energy utilization. Assuming the biosynthesis costs associated with net growth do not differ substantially from those associated with internal carbon turnover, this suggests that specific growth rate likely does not underestimate specific carbon turnover rate by more than ~30%. Given the likelihood that some energy is consumed in processes other than biosynthesis (e.g., [128]), the extent of the underestimate will be lower still.

Field Rates:

Pasture and forest soils [129]: Mass-specific carbon oxidation rates and specific growth rates are estimated from their Fig 3a and 3b, respectively. Mass-specific carbon oxidation rates are converted to MSP using a value of $\Delta G = -487 \text{ kJ (mol C)}^{-1}$ for terrestrial carbon (Supplementary Material Section 3.2).

Marine sediments [130]: Fig 3b of Lomstein et al. [130] plots paired values of biomass carbon residence time, T_b , for sediments ranging in depth from 1 to 40 meters below seafloor. One estimate of T_b was based on the observed extent of racemization in aspartic acid (here distinguished as T_{b-aa}), and is plotted in Fig. 4 as the measure of specific carbon turnover rate (where specific carbon turnover rate is the inverse of T_{b-aa}). In their Fig. 3b, each value of T_{b-aa} was paired with a corresponding estimate of T_b that was independently calculated from measured cell abundances and modeled sulfate reduction rates at the same depth (here distinguished as T_{b-sr}). T_{b-aa} and T_{b-sr} are based on completely independent measurements. Because their calculated values of T_{b-sr} were ultimately based on cell-specific sulfate reduction rates (volumetric sulfate reduction rate divided by volumetric cell abundance), they can be used as a basis for calculating MSP. In calculating T_{b-sr} from cell-specific sulfate reduction rates, Lomstein et al. [130] assumed that sulfate reducers comprise 13% of total cell abundance and have a growth yield of 0.11 gram biomass carbon per gram substrate carbon, and that the respiratory quotient (RQ: C_{org}/SO_4^{2-}) is 2. We used these same values to back-calculate mean cell-specific carbon oxidation rates from the values of T_{b-sr} plotted in their Fig. 3b. MSP was calculated from mean cell-specific carbon oxidation rates based on a mean cell mass of 14 fg C for cells in the marine deep biosphere [35] and a Gibbs energy change for sulfate-based respiration of organic carbon of $\Delta G = -32.2 \text{ kJ (mol C)}^{-1}$ (Supplementary Material Section 3.2).

4.2. Source data and MSP calculations for Figure 4

Aerobic Respiration: The plotted MSP is based on areal O_2 uptake rates in Aarhus Bay sediments of $1000 \mu\text{mol m}^{-2} \text{ h}^{-1}$ (September) and $250 \mu\text{mol m}^{-2} \text{ h}^{-1}$ (March) as reported in Rasmussen and Jørgensen [131]. Areal rates were converted to average volumetric O_2 consumption rates based on the measured depth of O_2 penetration (September: 0.15cm, March: 0.5cm), and then to average

volumetric C oxidation rates based on a respiratory quotient (RQ: C_{org}/O_2) of 0.90 [53]. Conversion of the bulk C oxidation rates to MSP assumes a Gibbs energy change for aerobic carbon oxidation of $-443 \text{ kJ (mol C)}^{-1}$ (Supplementary Material Section 3.2), a cell abundance in the aerobic sediment layer of $4 \times 10^9 \text{ cells cm}^{-3}$ (based on the reported values of $5 \times 10^9 \text{ cells cm}^{-3}$ [132] and $2 \times 10^9 \text{ cells cm}^{-3}$ [133]), and a mean cell mass in the aerobic layer of 23 fg C [44].

Sulfate-based Respiration: The plotted MSP is based on cell-specific sulfate reduction rates in Aarhus Bay sediments as reported in Petro et al. [132], Sahm et al. [134] and Holmkvist et al. [135]. Calculation of cell-specific power from cell-specific sulfate reduction rates assumes an energy yield to sulfate reducers of $DG = -27.4 \text{ kJ (mol SO}_4^{2-})^{-1}$. This value is based on a Gibbs energy change for sulfate-based carbon oxidation of $\Delta G = -32.2 \text{ kJ (mol C)}^{-1}$ (Supplementary Material Section 3.2) and a respiratory quotient (RQ: C_{org}/SO_4^{2-}) of 1.7 [74], yielding an overall Gibbs energy change for sulfate-based organic carbon respiration of $DG = -54.7 \text{ kJ (mol SO}_4^{2-})^{-1}$. As the overall oxidation of organic carbon to CO_2 involves both fermentative and sulfate-reducing bacteria, we assume that the total Gibbs energy change is shared evenly between the two groups. Thereby, sulfate-reducing bacteria realize an energy yield of $-54.7/2 = -27.4 \text{ kJ (mol SO}_4^{2-})^{-1}$. Calculation of MSP from cell-specific power is based on a mean cell mass of 23 fg C.

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